

t r u e

# HYBRID

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## **The Complete Framework**

Physical · Mental · Spiritual

*Capacity · Command · Direction*

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## PART I

# The Philosophy

*Capacity · Command · Direction*

## The thesis

A hybrid athlete is asked to be strong, muscular, and conditioned all at once. True Hybrid says that is only the first third of the problem. A complete athlete is built in three layers.

- **Physical, the capacity.** What the body can produce.
- **Mental, the command.** Whether that capacity shows up when it counts.
- **Spiritual, the direction.** What the whole thing is aimed at.

Three things hold those layers together as one model instead of three separate programs.

- **They run the same arc.** Regulate the base, build the capacity, then express it when it matters. The physical domains (Build, Endure, Express) are the template. The mental and spiritual layers run the same shape.
- **They share a nervous system.** Body and mind move together. Every change in your physical state changes your mental one, and it works the other way too. The same slow breathing that settles the body settles the mind. This is not a metaphor. The balance between your stress and recovery branches of the nervous system is the same balance that decides which muscle fibres you recruit and how composed you stay under pressure. Train one layer and ignore the others and you cap the whole thing.
- **Aiming up gives it a vector.** Capacity and composure still need somewhere to go. The spiritual layer points the system. Without it you are just working hard with no idea why.

***Build the base, earn the expression, aim it higher.***

### ON SOURCES

Almost none of the parts of this model were invented from nothing. The progression principles come from Pat Davidson. The contemporary, oxygen-led view of energy systems was inspired by the work of Evan Peikon. The conditioning methods grew out of a long lineage of physical preparation coaches. What is mine is the way it is all assembled: one system, tested and shaped over years of working with hundreds of real people, where the pieces are chosen because they are simple, true, and they hold up in application. Where a piece belongs to someone else, it is named where it appears.

## The hybrid problem, honestly

There is a real constraint underneath all of this, and we name it instead of pretending it away. Maximal force and deep endurance pull against each other. As the engine grows, peak force tends to give up ground, and it works the other way too. Past the beginner phase, nobody maximises both in the same window. That is not a flaw to coach around. It is the ground you are standing on.

So the job is not to train every quality equally and forever. The job is to put them in order. You give one quality the lead while the others sit on a maintenance dose, and you rotate

which one leads over time. And because the body does not behave like a machine, the same session gives different results in different people and different seasons, so the whole framework is steered by how the athlete responds, not by faith in a fixed plan.

That single discipline has a name we use all the way through this book.

### ***Bias the limitation, but train the full system.***

You never abandon a quality. You find the one thing holding the athlete back, give it the lead, and keep everything else ticking over so nothing slips while you fix it. Then you retest and rotate. The rest of this book is the machinery for doing exactly that.

## **Why hybrid breaks most programming**

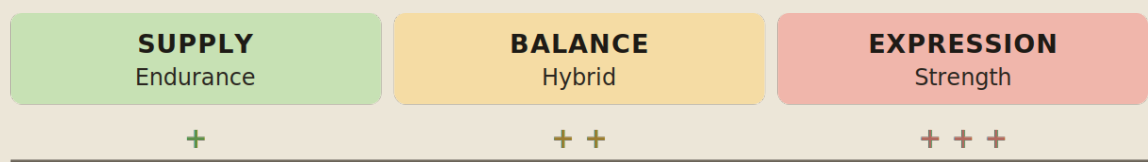
Training inherited two clean traditions. General physical preparation (GPP) builds the broad base. Specific physical preparation (SPP) sharpens the thing the sport actually demands. Most coaches run one or the other. The hybrid athlete breaks both, because the sport itself is no longer one thing. Read a modern mixed-modal athlete and the demand splits into three jobs that all have to happen at once.

| The demand          | What it asks of the athlete                                            |
|---------------------|------------------------------------------------------------------------|
| <b>Conditioning</b> | The engine. Supply, and the ability to repeat output across durations. |
| <b>Skill</b>        | Body control and the ability to express force through good position.   |
| <b>Lifting</b>      | Maximal and near-maximal force. The top end of expression.             |

Train these as three separate programs and they fight each other. Train them as one undifferentiated mess and none of them improves. The way through is to see that all three sit on the same two underlying spectrums, what the athlete is and where on the engine they are working, and to program against those rather than the surface activity.

## **The three athletes**

Every athlete sits somewhere on a single line that runs from pure supply to pure expression. We mark three reference points along it. Most real athletes live between them and always lean one way.



*The supply-to-expression spectrum. Supply is the heart and lungs, the delivery side. Expression is the muscular system, the output side. Balance is the hybrid in the middle, needing enough of both.*

| Archetype        | Internal state | Tension / skill | Leans toward                                                      |
|------------------|----------------|-----------------|-------------------------------------------------------------------|
| <b>Endurance</b> | Supply         | Low ( + )       | Delivery. Moving oxygen and fuel, over and over, for a long time. |
| <b>Hybrid</b>    | Balance        | Moderate ( ++ ) | Both at once, which is why it carries                             |

| Archetype | Internal state | Tension / skill | Leans toward                                        |
|-----------|----------------|-----------------|-----------------------------------------------------|
|           |                | )               | the most conflict from interference.                |
| Strength  | Expression     | High ( +++ )    | Producing force. The muscular expression of output. |

Keep this line in mind for the whole physical layer. Everything that follows, the domains, the tests, the limiter logic, is a way of finding where an athlete sits on it and then biasing the gap.

## PART II

# The Physical Layer

## *Capacity, what the body can produce*

This is the engine room. Part I gave you the why. This part builds the full picture of the physical system and the language we use to run it. How we define fitness, the two things that shape every athlete, the oxygen-led engine underneath, and the Build, Endure, Express domains that organise the work.

## Defining fitness

Fitness is not one number. It is a profile across several qualities, and an athlete is just the particular shape of that profile. We read it on one instrument we call the Fitness Compass. The qualities fall into two groups, and the groups do different jobs.

- **Descriptive qualities** tell you what state the system is in right now. They describe the athlete. CP/CS (critical power or critical speed), VO2, HRV (heart rate variability), and RHR (resting heart rate).
- **Prescriptive qualities** are outputs you can measure today and write straight into a session tomorrow. Strength, power, speed, and mobility.

Read the two groups as two separate lists, not as matched pairs. HRV does not pair with speed, and RHR does not pair with mobility. Each group is a set of dials you read together. A complete read needs the whole compass, the state on one side and the output on the other, not one favourite number.

| DESCRIPTIVE | <i>the state the system is in now</i> | PRESCRIPTIVE | <i>outputs you program tomorrow</i> |
|-------------|---------------------------------------|--------------|-------------------------------------|
| CP / CS     | sustainable power or speed            | Strength     | force you can produce               |
| VO2         | aerobic ceiling                       | Power        | force at speed                      |
| HRV         | recovery and readiness                | Speed        | top-end velocity                    |
| RHR         | baseline stress                       | Mobility     | usable range                        |

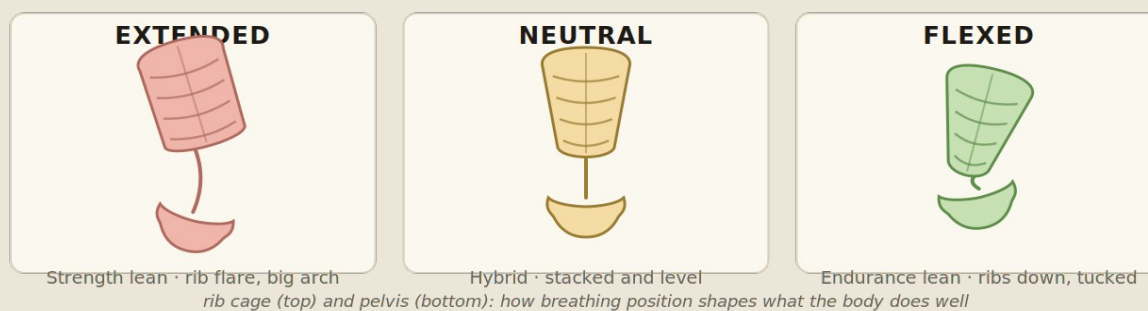
*The Fitness Compass. The descriptive side reads the state the system is in, the prescriptive side reads outputs you can program straight into a session. Read them as two lists, not matched pairs.*

## Structure and function

Conditioning does not happen in a vacuum. Every performance comes out of the athlete, the task, and the environment interacting. The part we can coach most directly is the athlete, and the athlete is two things: structure and function.

## Structure dictates function

Structure is the athlete's architecture. The clearest version of it is the relationship between the rib cage, the diaphragm, and the pelvis, and the breathing position that relationship creates. This is not cosmetic. It decides what the body is good at.



*Three structures. The strength lean carries an extended, rib-flared position built to brace and produce force. The endurance lean carries a flexed, ribs-down position built to breathe and absorb. The hybrid sits near neutral, stacked, and can lean either way.*

An endurance athlete tends to stack the rib cage over the pelvis with a long, usable breathing position, built to take air in and deliver it. A strength athlete tends to carry a more extended, braced position built to lock down and push force. The hybrid sits closer to neutral but is always leaning one way. You cannot recruit a muscle you cannot get into position, and breathing is the first thing that falls apart when position is wrong. So structure decides function.

### Function over time changes structure

The arrow also runs the other way, slowly. What you repeatedly ask the body to do remodels the architecture. Tissue, position, and tolerance all adapt over months to the work you give them. This is why the movement and breathing work later in the book is not mobility for its own sake. It is the slow editing of structure so that better function becomes available.

Take a real example. A powerlifter has spent years bracing hard against heavy bars. Their structure has set into extension: rib cage flared up and forward, lower back compressed, pelvis tipped forward. That is a brilliant structure for a maximal squat and a poor one for distance running, where you need to get air in and out efficiently for an hour. If that lifter now wants to become an endurance athlete, you cannot just hand them a running program. You have to change the structure first, and you do it with exercise selection that pulls them out of extension and into a stacked, flexed-capable position.

- **Get the ribs down over the pelvis.** Exhale-biased work where you fully blow air out and feel the front of the ribs draw down. Things like 90/90 breathing with the hips and ribs stacked, dead bug variations with a full exhale, and back-supported breathing drills.
- **Train positions, not just muscles.** Choose exercises that reward a stacked position and punish rib flare. Hands-elevated or supported work first, holding a long exhale at the top, before loading the same pattern free and heavy.
- **Bias the opposite of their default.** A braced extender needs reaching, rounding, and breathing under light load. Think exhale-led carries, controlled flexion patterns, and posterior expansion drills, where someone built for endurance who wants to get strong needs the reverse: bracing, stiffness, and tension they currently cannot produce.

This takes months, not weeks, and it runs alongside the conditioning, not before it. But without it the running never feels right, because the structure underneath is still built for a different job.

## So what is function?

We use a precise definition and keep it. Function is the combination of energy systems and muscle fibres, organised for a task. Not one energy system, not one fibre type, the combination. That definition is the hinge between the structure work above and the energetics below, and it is what the domains later in this part are organising.

## Skill, tension, and the nervous system

Skill is easy to leave out of a physical model because it feels like a separate topic. It is not. Skill changes how much tension a movement costs and how hard the nervous system has to work, so it belongs right next to strength and conditioning.

A low-skill athlete produces tension everywhere. They brace muscles that should be relaxed, fire antagonists against the movement, and waste effort fighting themselves. That wasted tension is expensive. It drives the heart rate up, drains the battery faster, and fatigues the system long before the working muscles are actually done. A high-skill athlete produces tension only where and when it is needed, and relaxes everything else. Same external output, far lower internal cost.

That is why skill sits on the same low-to-high line as tension. As you climb from Build to Express, you are not only asking for more tension, you are asking for more skill to control it. A slow bike ride is low skill and low tension. A heavy, fast, technical lift under fatigue is high skill and high tension, and the nervous system has to both produce force and coordinate it at speed. Skill is what lets an athlete spend tension efficiently, which means it directly sets how much real work they can do before fatigue ends the session. Build skill and you lower the nervous-system cost of everything above it.

## Supply, expression, and what to actually build

The supply and expression line from Part I is physiological, not just a label.

- **Supply** is the heart and lungs. The delivery system. The endurance athlete lives here.
- **Expression** is the muscular system. The producer of output. The strength athlete lives here.
- **Balance** is the hybrid. Enough of both, always leaning, and the athlete who pays the highest interference tax for trying to hold the middle.

## The headline and the support

Here is the idea that stops a lot of wasted training, and it is worth being very plain about because it is easy to read backwards. For any athlete, one quality is the headline and the other is the support. The headline is the one you bias and keep building. The support only needs to be good enough for the sport, and once it is good enough, you stop pushing it.

Take a strength athlete. Their headline is expression, so that is what you bias. They also need some supply, enough aerobic function to recover between heavy efforts, but that is the support. It does not mean you fill their week with conditioning. It means you give them just enough conditioning to be sufficient for their sport, and then you stop and put the budget back into expression. Reading it the other way, that a strength athlete needs more and more supply work, is exactly the mistake to avoid. More support past sufficient is wasted training that eats into the headline.



## The sufficiency principle

Said as a rule: several of an athlete's qualities only need to be sufficient, not maximal. VO2 and expression are often in this group. A strength athlete needs enough aerobic function to recover and enough expression to meet the task, not the most either can be. Chasing the maximum of a quality that only needs to be sufficient is how someone spends a whole block and moves nothing that mattered. Ask what the sport actually requires, satisfy it, and spend the rest of the budget on the real limiter.

## The engine, an oxygen-led model

The old picture draws three energy systems as separate boxes, alactic, anaerobic-lactic, aerobic, handing off to each other one at a time. The view we use, inspired by the work of Evan Peikon, is that they overlap and run together, and that one thing connects all of them.

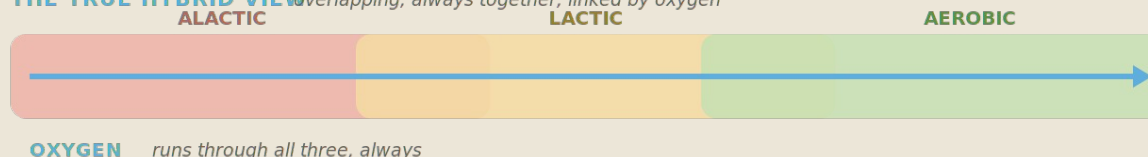
**Oxygen is the link. It is always there, and it is always needed.**

Even the work we call anaerobic depends on oxygen the moment you look at recovery and at rebuilding for the next effort. So instead of asking which system, we follow the oxygen. Into the muscle, through the effort, and back out. That thread never breaks, and it keeps the whole model simple.

**THE OLD VIEW** *three separate systems, handing off one at a time*



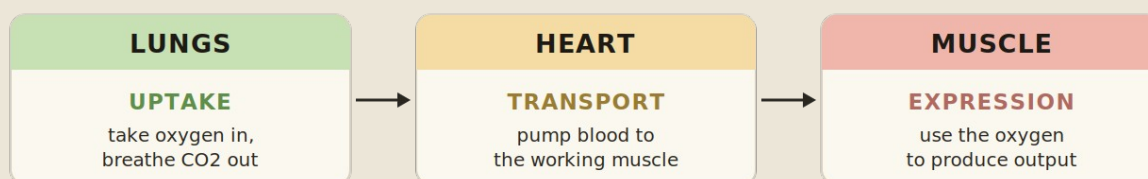
**THE TRUE HYBRID VIEW** *overlapping, always together, linked by oxygen*



*The old view splits the engine into three separate systems that hand off one at a time. The True Hybrid view runs them together, overlapping, with oxygen threading through all three at once.*

## Follow the oxygen, the broad view

Zoom out and the body is a supply chain with three links. Each link is an organ system doing one job. Read the words carefully, because uptake, transport, and expression belong here, to the cascade. They are not the names of the training domains.



*The oxygen cascade. The lungs take oxygen in. The heart moves it to the working muscle. The muscle uses it to produce output. Every limiter you will ever chase sits somewhere on this chain.*

The full loop, breath by breath: you breathe in, the blood carries oxygen to the muscle, the muscle uses it and produces carbon dioxide, the blood carries that back, and you breathe it out. When you go looking for what is holding an athlete back, you are looking for the weakest link on this chain. Testing later in the book is how you find it.

### What is happening when output drops

At the working muscle, output is limited by how much oxygen is actually available right there in the tissue. As the local oxygen gets used up, the muscle can no longer keep contracting at the same level, and output falls. Three local things drive that drop: the muscle squeezing its own blood vessels as it contracts, which chokes off flow; blood struggling to get in and backing up on the way out, which slows both delivery and clearance; and the muscle itself burning through oxygen faster than it arrives, so its own demand outruns the supply.

#### CAVEAT

You do not need to measure any of this, and most people never will. There is a tool that reads muscle oxygen directly, but you do not need it and you do not need to memorise the terminology. What matters is the understanding. When you know that output drops because the muscle has run low on oxygen, you know what the limitation actually is, and that tells you what to do about it: improve delivery, improve the muscle's ability to use what arrives, or manage the tension that is choking off its own blood supply. How you tell which of those three is the limiter is set out in Part VII under Assess. The molecular picture is there to guide the decision, not to be tested.

### Muscle fibres

Three fibre types, brought in as the demand rises.

- **Type 1, slow and aerobic.** The supply end. Fatigue-resistant, recruited first.
- **Type 2A, fast but still aerobic.** The middle ground.
- **Type 2B, fast and glycolytic.** The expression end. High force, fast to fatigue.

They are recruited in order. As force goes up, the body brings in progressively higher fibres, Type 1, then 2A, then 2B. This is the reason behind a rule we come back to later: when a muscle will not grow, the first question is whether you are even reaching the right fibres, not whether to add more sets. You have to recruit a fibre before you can load it.

### The shared nervous system

The same nervous-system state that sets your fibre recruitment also sets your mind. This is the real wiring behind the shared-nervous-system idea, and it is the handshake between this part and the mental layer.

| Nervous-system state             | Fibre bias                   | How it shows up                                                       |
|----------------------------------|------------------------------|-----------------------------------------------------------------------|
| <b>Parasympathetic (recover)</b> | Type 1, slow and aerobic     | Settled, recoverable, durable, composed.                              |
| <b>Sympathetic (stress)</b>      | Type 2B, fast and glycolytic | Switched on, high output, expensive, fragile if you do not manage it. |

Breathing is the lever that moves both layers because it is the one input that shifts this balance on command. The same slow breathing that pushes you toward recovery and Type 1 function is the breathing that gives you composure under load. One nervous system, trained from two directions.

### The levers on each link

Each link of the chain has its own set of things you can change to improve it. This is the level of detail you reach for when you have found the limiter and need to know what to actually do.

| Link          | What you can change to improve it                                                            |
|---------------|----------------------------------------------------------------------------------------------|
| <b>Lungs</b>  | Nervous system, the breathing muscles, and the structure of the rib cage and trunk.          |
| <b>Heart</b>  | Nervous system, the circulatory system, and the heart itself.                                |
| <b>Muscle</b> | Nervous system, fibre type, blood flow, and structure, and tension runs through all of them. |

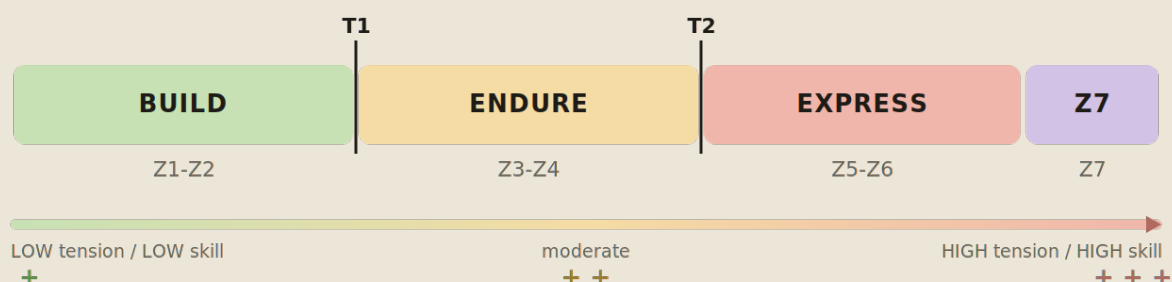
Tension shows up in every muscle lever, which is one reason it is one of the two things we never stop watching. The same tension that builds muscle also squeezes off its own blood supply and adds to fatigue. Manage it well and it works for you. Manage it badly and it quietly wrecks concurrent training.

#### CAVEAT

The limiter can sit at either end. This model reads the muscle end closely, and that is on purpose, because it is where most of the practical levers are. But supply is genuinely two-sided. Central means the heart and lungs cannot deliver enough oxygen, so the whole system runs out at once. Peripheral means delivery is fine but the muscle cannot extract and use the oxygen fast enough, so it fails locally. Most athletes are limited one way or the other at any given time, and the point of testing is to find out which, not to assume. How to tell them apart, central, peripheral, or tension, is set out in Part VII under Assess.

### The domains, Build, Endure, Express

Now the spine. The engine runs across one line of demand, from easy and sustainable at one end to maximal and brief at the other. We split that line with two thresholds, T1 and T2, into three domains. These are the names we use everywhere in the program, and they are not the cascade words above.



The domains across one line. Build (Z1-Z2) sits below T1. Endure (Z3-Z4) sits between T1 and T2. Express (Z5-Z6, with Z7 above) sits past T2. The same line is also a low-to-high tension and skill gradient.

| Domain         | Zones | What it does                                                                                     | Skill level, with examples                                                                                                           |
|----------------|-------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>BUILD</b>   | Z1-Z2 | Deliver. Move blood, oxygen and fuel efficiently and repeatably. Easy, sustainable, low cost.    | Low skill. Steady bike, easy row, slow air squats, a long easy run, sled drags. Simple, repeatable, hard to get wrong.               |
| <b>ENDURE</b>  | Z3-Z4 | Hold output as the cost climbs. Work near the edge of what you can sustain. Moderate everything. | Moderate skill. Threshold intervals, loaded carries, moderate-load circuits, tempo work, a hard-but-controlled CrossFit-style piece. |
| <b>EXPRESS</b> | Z5-Z6 | Produce output above what delivery can sustain. Short, expensive, maximal.                       | High skill. Heavy and fast lifts, jumps and bounds, sprints, complex barbell work done with quality under fatigue.                   |
| <b>(Z7)</b>    | Z7    | The very top. Pure speed and power, almost no repeatability.                                     | Highest skill. Maximal sprints, single big jumps, true max-effort efforts with full recovery.                                        |

Those two thresholds are not arbitrary lines, they are the two points where the body's metabolic state actually changes, which is why we build the whole spine around them. T1, the first threshold, sits between Z2 and Z3. Below it you are fully aerobic: lactate stays at its resting baseline, you are burning a lot of fat, breathing is easy, and you could hold the effort for hours. Cross T1 and lactate starts to rise above baseline, fuel shifts toward carbohydrate, and breathing deepens. You are now working, not cruising. T2, the second threshold, sits between Z4 and Z5. It is the highest intensity at which you can still clear lactate as fast as you make it, the top of metabolic steady state. Below T2 you are holding a balance, hard but sustainable for a long stretch. Cross T2 and the balance breaks: lactate and metabolites pile up faster than you can clear them, breathing becomes maximal, and you are on a clock measured in minutes, then seconds. So the three domains are really three metabolic states. Build sits below T1 where everything is sustainable and cheap. Endure sits between the thresholds where you hold the edge. Express sits past T2 where steady state is gone and output is bought with time you do not have.

Read the same line as a skill gradient and it tells you something the table makes concrete: the further right you go, the more skill the work demands, because you are asking for more tension and more coordination at higher speed. You build cheaply in Build, you do the real working volume in Endure, and you earn Express. You do not start an athlete in Express. The engine and the connective tissue have to earn the right to express, and so does the skill.

#### CAVEAT

Z5 lives in Express. T2 falls between Z4 and Z5, so Z5 is the bottom of Express, not the top of Endure.

#### UNDER THE HOOD

How the zones are anchored. To make the zones real rather than a vague feeling, we tie them to Critical Power. Critical Power is the hardest effort you can hold for a long time

without falling apart, a pace you could sustain for a long stretch but not push much past. Below it, you are comfortable. Right around it, you are at the edge. Above it, you are spending a reserve that runs out fast, like a small extra tank you can only use for short bursts before it is empty and needs refilling. Build sits well below Critical Power. Endure sits around it. Express sits above it, dipping into that reserve. Testing later shows the simple two-effort protocol that finds an athlete's Critical Power and the size of their reserve.

## Reading the domains in halves

Each domain splits into a low and a high shade, and the split is not cosmetic. Within a single domain, two things shift as you move from the low half to the high half: what your metabolism is actually doing, and how long the work takes to recover from. Most of the time you steer by the three domains. When you want to be precise about a limiter, is this athlete's engine failing at threshold or at VO<sub>2</sub>, and how much recovery does that cost me in the week, you reach for the half.

This table is also the translator. When an older program or training sheet hands you an effort percentage or a heart-rate zone, this is where it lands. Lead by RPE for conditioning as always, and treat the effort and heart-rate figures as the way to pin a shade down when you are biasing that quality hard.

| Shade               | RPE  | % effort | % max HR | Lactate                                                  | Recovery before you repeat it                                            |
|---------------------|------|----------|----------|----------------------------------------------------------|--------------------------------------------------------------------------|
| <b>Build low</b>    | 3-4  | ~65%     | 50-65%   | At resting baseline, none above it.                      | None. Train daily, twice a day, or back to back with no cost.            |
| <b>Build high</b>   | 4-5  | 70-80%   | 65-75%   | Very little above baseline.                              | Low. Multiple sessions in 24 hours and back-to-back days are fine.       |
| <b>Endure low</b>   | 6-7  | 80-85%   | 75-85%   | Little above baseline.                                   | Moderate. Inside 24 hours is fine, but not smashed daily.                |
| <b>Endure high</b>  | 7-8  | 85-90%   | 85-90%   | Some, expected, especially near the top.                 | About 48 hours between hard sessions.                                    |
| <b>Express low</b>  | 9    | 90-95%   | 90-95%   | Accumulates, but should not limit pace until very late.  | About 72 hours.                                                          |
| <b>Express high</b> | 9-10 | 95-100%  | 95-100%  | Produced heavily; you hold intensity as acidosis climbs. | At least 72 hours. The greatest recovery demand of anything you program. |
| <b>Z7</b>           | Max  | ~100%    | n/a      | Produced, but fully cleared between reps.                | Low systemic cost despite the intensity, because the bouts               |

| Shade | RPE | % effort | % max HR | Lactate | Recovery before you repeat it      |
|-------|-----|----------|----------|---------|------------------------------------|
|       |     |          |          |         | are tiny and the rest is complete. |

Read the lactate column top to bottom and it tells one continuous story, which is the point of the whole model. In Build you are at or barely above resting baseline, which is why it is nearly free and you can stack it. Cross into Endure and lactate starts to appear and must be cleared, which is why the recovery clock starts ticking. In Express low it is piling up but you can still hold pace almost to the end; in Express high it is the thing you are fighting, and holding output against rising acidosis is the entire stimulus. The three metabolic states from the domains section are really six, and the recovery cost roughly doubles at each step down the table.

#### UNDER THE HOOD — YOU HAVE TO BE STRONG ENOUGH TO EARN THE TOP

Express high is the one shade an athlete can be too underdeveloped to train. Give a weak or underpowered athlete a maximal lactic-power session and they cannot produce enough output to actually load the zone. They get a poor stimulus and a large recovery bill for it, the worst trade in the book. The fix is the same order the whole model runs on: build absolute strength and the alactic top (Z7) first, so the athlete can produce the output that Express high demands, and only then bias Express high itself. This is earn the expression made literal. When the top end will not improve, the first question is not more intervals, it is whether the athlete is powerful enough to reach the zone at all.

#### CAVEAT — Z7 DOES NOT FOLD INTO EXPRESS HIGH

The pure alactic top is a seventh shade that sits above the six, not inside them, because its recovery profile is the opposite of Express high. Short maximal bouts with complete rest, lactate produced but fully cleared before the next rep, heart rate not a useful metric, and a low systemic cost despite the intensity, hard but comfortable, because the rest is total. Express high is maximal effort you pay dearly for; Z7 is maximal effort you barely pay for. Keep them apart, and keep Z7 where the domains section already has it: the bracketed top of the line.

### The halves as a progression and as different adaptations

The low and high shade of each domain are not just two intensities. They are two tools, and they do two separate jobs.

First, they are a **progression**. Within any domain you build the low half before you load the high half. Build low, big, cheap, sustainable aerobic volume, is the foundation you lay before Build high asks the same system to work a little harder. Endure low, the recoverable, near-threshold end, is where you earn the right to Endure high at true threshold. Express low, driving the aerobic ceiling, comes before Express high, holding output against rising acidosis. This is the same logic as the contraction line and the movement map: own the cheaper, more repeatable version first, then progress into the more expensive one. An athlete with no Build base has no business living in Endure high, and an athlete who cannot hold Express low has nothing to express in Express high. You walk up the halves the way you walk up everything else.

Second, and just as important, the two halves **train different adaptations**, so the split is also a way of aiming at a specific limiter rather than just turning up the effort.

| Shade               | What it develops                                                                                                                                      |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Build low</b>    | The delivery base, cardiac output efficiency, capillary density, fat oxidation, and the slow-twitch, aerobic machinery. Durability at almost no cost. |
| <b>Build high</b>   | The same base worked a shade harder, shifting fuel toward carbohydrate and building muscular endurance underneath the engine.                         |
| <b>Endure low</b>   | Pacing, heart-rate recoverability, and the bridge into intensity, the ability to surge and settle.                                                    |
| <b>Endure high</b>  | Threshold: raising the ceiling at which you can still clear lactate as fast as you make it. The top of sustainable output.                            |
| <b>Express low</b>  | The aerobic ceiling itself, VO <sub>2</sub> , central delivery, driving the heart to its peak.                                                        |
| <b>Express high</b> | Lactate tolerance and glycolytic power, producing and holding output while acidosis climbs.                                                           |

Read the two readings together and the practical move falls out. When an athlete is early, or under-based, or the limiter is delivery, you use the split as a progression and spend your time in the low halves. When the base is already there and you have found a specific limiter, threshold, or VO<sub>2</sub>, or lactate tolerance, you use the split as an adaptation target and bias the exact half that trains it, holding the rest. Same six shades, used one way to build an athlete up the line and another way to attack a named gap. Which one you are doing depends, as always, on where the athlete sits and what the assessment says is missing.

## The five dials, and when to reach for each

You can put a number on effort five ways, and the whole trick is knowing which dial to reach for. Two of them are about feel and three are about precision.

- **RPE**, how hard a whole piece feels out of ten. Your default for conditioning.
- **RIR**, reps in reserve, how many reps you had left on a lift. Your default for lifting.
- **Watts**, read off your Critical Power test. Precision for cyclical conditioning on the bike, row, or ski, where the machine gives you a clean number.
- **Run pace**, minutes per kilometre off your critical speed or threshold-pace test. This is the running equivalent of watts. Reach for it when the work is running, where there is no power meter, and you want to hold an exact pace rather than guess by feel.
- **Percent of one-rep max**, or percent of max heart rate. Precision for a specific load or a specific heart-rate zone.

Here is the rule that ties them together, because the rest of the book leans on it. RPE and RIR are the dials you steer by day to day. They are the only ones that work cleanly in a mixed piece, where a session blends running, lifting, and skill work and there is no single clean watt or pace to read. They are also the right tool whenever you are maintaining a quality rather than chasing it. The precise dials, watts, run pace, percent of max, come out when you are developing a specific quality and need to hit an exact dose: biasing strength to a true heavy load, or holding an aerobic pace to the watt so you know the engine is getting the right stimulus. So the rule of thumb is simple. When you are biasing something and precision matters, get specific. When you are maintaining, or running mixed work, run on



feel. Most of your week runs on RPE and RIR. The precise dials are for the one or two things you are deliberately pushing.

When an older prescription hands you an effort percentage instead of an RPE, translate it through the shade table in Reading the domains in halves rather than guessing, then steer by feel from there.

Tie the feel dials to the domains so you always have a target, even with no meter in sight.

| Domain         | RPE (out of 10) | RIR on lifts  | Feels like                                              |
|----------------|-----------------|---------------|---------------------------------------------------------|
| <b>Build</b>   | 3-5             | 4+ reps left  | Easy, conversational, could keep going for a long time. |
| <b>Endure</b>  | 6-8             | 2-3 reps left | Working, uncomfortable, holding the edge on purpose.    |
| <b>Express</b> | 9-10            | 0-1 reps left | Maximal or near it, short, full recovery between.       |

#### APPLY IT

Take one of your own training pieces from this week. Decide which domain it actually belongs in by feel, then put a number on it: an RPE for the whole piece, and an RIR if it has lifting in it. If the piece was meant to be a Build session but it kept creeping to an RPE of 8, it was not a Build session. That mismatch, the gap between what you intended and what you did, is one of the most common reasons training stalls. Get in the habit of naming the domain and the number before you start, not after.

## The contraction axis

The domains answer how hard and how long. Bolted onto them is a second line that answers how the muscle is asked to contract, and the two are lined up on purpose. The contraction line runs from owning tension, to holding position, to controlling tension under load, to releasing it at speed. We line it up with the domains because the right to express force fast is something the muscle and the connective tissue have to earn first.



*The contraction line, aligned to the domains. You own the shortening, then hold position, then control the lowering, then release at speed. Each stage earns the next.*

## The domains are a progression, not separate boxes

This is the part people miss. Build, Endure, and Express are not three different exercise pools. They are the same movements made progressively harder. Everything you own in Build can be carried forward into Endure, just heavier, more complex, and with more tension, and everything you own in Endure carries forward into Express, heavier and faster again.

Take a squat. In Build it is a slow, controlled goblet or tempo squat at a light load, where the job is clean reps and a big base with low cost. Move it into Endure and it becomes a heavier



front squat, a tempo back squat, or loaded squatting inside a mixed piece, where the contraction is more demanding and the tension is higher. Move it into Express and it becomes a heavy, fast squat, a jump squat, or a clean, where you are now releasing force at speed. Same pattern, walked up the line. A carry follows the same path: a light farmer carry in Build, a heavy or odd-object carry inside an Endure piece, a short maximal yoke or sprint-loaded carry in Express. When you progress an athlete from one domain to the next, you are not throwing out what they did, you are loading and complicating the things they already own.

This is the practical version of the contraction line above. As a movement climbs the domains, the contraction it asks for climbs too: from owning the shortening, to holding and controlling tension, to releasing it fast. Pick the version of the movement that matches the domain you are training, and progress the movement and the domain together.

### **Skill and load sit on the axis too**

The contraction line is not only about the type of contraction. It also carries skill and load, and those can swing hugely inside a single stage. Take concentric work. A light, continuous leg extension and a heavy walking lunge are both concentric, but they are worlds apart: the leg extension is low tension, low skill, and endlessly repeatable, while the walking lunge is higher tension, asks for real balance and coordination, and cannot be repeated anywhere near as often. Same contraction, very different demand. So naming the contraction is only the start. You also choose how much skill and how much load the movement carries, and you climb those the same way you climb the domains: low and repeatable first, higher and more demanding as the athlete earns it.

Load is the dial that decides most of this, and it is the last one you turn up, not the first. Concentric base work has to stay light and repeatable, because the whole point is to accumulate quality volume without a recovery bill. What counts as light is entirely athlete-dependent: a load one person waves off for sets all week is a genuine session for another, so you set it by what they can repeat with clean form at low cost, not by a number on the bar. As a movement climbs the line, the load goes up and the repeatability comes down. The heavier and more skilled the contraction, the more it costs and the less of it you can do. Match the load to the job, keep the base truly light, and spend the heavy, expensive, less-repeatable work only where you mean to.

This is not abstract. The contraction type changes which exercises you pick, and picking the wrong one is how athletes get hurt or stall. Here is each stage with the kind of movements that belong in it and why.

### **Concentric, own the shortening (Build)**

This is the contraction you build volume with, where the muscle shortens to move the load. It is repeatable and relatively cheap on the tissue, which is why it sits at the base. Pick movements where you can accumulate quality reps without paying a big recovery bill: sled pushes and drags (almost pure concentric, very little soreness), prowler work, concentric-only or short-range machine work, cycling and the assault bike, and lighter loaded carries. When the goal is to add work to the base without wrecking the legs for tomorrow, this is where you live.

**Isometric, hold the position (Endure, entry)**

Here the muscle produces tension without moving, holding a position against a load. This is the safest way to start loading tendons and to teach a joint to tolerate a position before you ask it to move under load. It is the entry point into the middle of the model. Good choices: wall sits and split-squat holds, mid-range pause work in a squat or bench, planks and loaded carries for the trunk, the Copenhagen and side-plank family for the hips, and tendon-focused holds like a long mid-range calf-raise hold or a heavy isometric pull against pins. Use these deliberately when a tendon is the weak link, because tendons adapt slower than muscle and respond well to time under tension with little movement.

**Tempo eccentric, control the lowering (Endure, loaded)**

This is the controlled, slow lowering where the muscle stays loaded and full of tension the whole way down and never gets to relax. It is expensive and it demands control, which is why it sits in the middle and not at the base. Be precise about it: a three to five second lowering on a squat, bench, or RDL, slow eccentric chin-ups and dips, tempo split squats, and slow lowering on machine work for muscles you are trying to grow. This is a powerful tool for building muscle and toughening tissue, but it produces real soreness, so you place it where it will not poison the next priority session. Do not stack heavy tempo eccentrics for the legs the day before a hard run.

**Reactive, release at speed (Express)**

The top of the line, where the muscle stretches and then fires back fast, storing and returning energy like a spring. High velocity, high skill, and only safe once the stages below are in place. Choices here: pogo hops and ankle bounces to start, then box jumps and depth jumps, bounding and broad jumps, medicine-ball throws, Olympic lifts and their variations, and fast, light barbell work where the intent is speed. You earn your way into this. A reactive, single-leg, change-of-direction movement is not something you hand an athlete whose base is still slow and supported.

**CAVEAT**

Name the two eccentrics every time, because the word hides two different tools. A fast, passive eccentric is cheap and springy, and that lives at the top as the reactive, spring-like work. A slow, controlled tempo eccentric holds high tension the whole way down, and that is expensive and lives in the middle. They are not the same thing, so say which one you mean.

**CAVEAT**

The base is not purely concentric. Calling Build concentric-led describes the tools you reach for there, like the sled and the bike, not a claim that nothing lengthens. Running is heavily eccentric every time the foot hits the ground. If your base conditioning is running, you are already banking a lot of eccentric load, so account for it before you pile tempo leg work on the same day.

**APPLY IT**

Look at your current leg day and label every exercise by contraction type. Most people find they are stacked at one end, all concentric machine work, or all heavy tempo eccentrics, and wonder why they are either not progressing or always sore. Rebalance it: a concentric base movement, an isometric for a joint or tendon that needs it, one tempo

eccentric for the muscle you most want to grow, and reactive work only if you have earned it. Then check it against your week. The reactive and tempo work should not be sitting the day before your hardest priority session.

### Breaking the rules: running in Build

These are defaults, not laws, and the useful skill is knowing when to break them for the athlete in front of you. Running is the clearest case. It is genuinely good Build work and a big aerobic stimulus, but it is a high-cost contraction: every footstrike is a hard eccentric load, far more punishing on the legs than the same aerobic minutes on a bike. So the question is never only whether running counts as Build. It is who you are giving it to.

Give a heavy powerlifter with no running base a block of Build running and you will bury them. They are not adapted to the eccentric cost or the movement, and the bill lands on the exact legs they lift with. For them you build the engine on the bike, the row, and the ski first, and progress toward running slowly, letting the tissue and the energetic systems adapt to the load and the pattern before you lean on it. Someone who already runs is the opposite case: they are adapted, so running belongs in their Build straight away. Either way the point is the same. Know the cost of the contraction relative to a bike, so you know how much of it the athlete can take and what to pair it with. Break the rule on purpose, with your eyes open to the cost, not by accident.

### Hypertrophy and strength inside the domains

The resistance qualities ride the same spine. A few claims everything else leans on.

- **Tension drives growth.** How close you get to failure and how much continuous tension you hold matter more than any magic rep range. Anywhere from roughly 30 to 90 percent of your one-rep max can build muscle if you take it close enough to failure, with the bottom cutoff around 20 to 30 percent. A slow tempo eccentric reads as more than a normal rep at the same load because it keeps the muscle under tension longer and adds metabolic stress on top.
- **The dose, and keep the two numbers separate.** Roughly 8 to 14 hard sets per muscle per session is the working ceiling. Weekly volume is higher and split across sessions, and higher weekly volumes do better with more frequency. Per session and per week are different numbers, so never quote them as one.
- **Recruitment before volume.** When a muscle will not grow, more sets is usually the wrong answer. Check whether you are recruiting it at all, whether the position is right, whether the execution is clean, and whether recovery is there, before you add work. This is where the structure and breathing work earns its place. You cannot recruit what you cannot get into position.
- **Strength is more than size.** It is also nervous-system drive, technique, arousal, and the energy system feeding the effort. A heavy endurance block can look like a strength loss that is partly the nervous system handling tension differently, not lost muscle.
- **Mind the connective tissue.** Tendons, ligaments, and bone are the tissues hybrid loading actually injures, and they adapt slower than muscle. The isometric work in Endure is the safe way to load them, so use it on purpose.

## Loading connective tissue

Tendons, ligaments, and bone are the tissues hybrid loading actually injures, and they are the most ignored. They adapt slower than muscle, on a timeline of weeks to months rather than days, and the classic hybrid injury is muscle and nervous system getting strong faster than the connective tissue can keep up. When your strength outruns your tendons, something tears. So you load tissue on purpose, early, and patiently.

### Isometrics, the entry point

Long, hard holds are the safest and most effective way to start loading a tendon. Hold a demanding mid-range position against a heavy load for thirty to forty-five seconds, three to five sets, three or four times a week. A heavy mid-range hold in a squat or a leg extension for the knee, a long calf-raise hold for the achilles, a mid-range isometric pull for a stubborn tendon. This builds tendon stiffness and tolerance, and it also calms cranky tendons down, which is why it doubles as a way to train through low-level tendon pain rather than around it. This is the isometric work that sits at the entry to the Endure domain, and it is where tissue loading should begin.

### Heavy slow resistance, the next step

Once a tendon tolerates isometrics, progress to heavy slow resistance: a moderate to heavy load lifted and lowered slowly, around three seconds up and three seconds down, for several sets of six to ten reps. This keeps the tissue under long, controlled tension through a range and is one of the best-evidenced ways to remodel a tendon and make it more robust. It maps onto the tempo eccentric work in the middle of the model.

### Then, and only then, speed

Reactive, fast, spring-like loading is the most demanding thing you can ask of a tendon, and it belongs last. You earn it with the isometric and slow work first. Rush an athlete into jumping and bounding before the tissue is ready and you get exactly the injury the model is built to avoid. This is the same logic as the contraction line: own tension, control tension, then release it.

### Bone, and the thing underneath all of it

Bone responds to load and impact with variety and progressive overload, the same as everything else, but it has one hard requirement: it will not build, and will actively weaken, if the athlete is chronically under-fuelled. Low energy availability erodes bone directly. So the connective tissue plan is only as good as the fuelling underneath it. Load it patiently, progress it from holds to slow tempo to speed, and feed it, or none of it sticks.

## Combining them, the hybrid problem in practice

Running strength, size, and conditioning together without each one blunting the others is a problem of order, modality, and dose.

- **Order.** The priority quality goes first, in the session and in the week, while you are fresh. Fast, high-skill, nervous-system work leads. Fatiguing work follows. Do not pour maximal power and deep fatigue into the same session and expect both to adapt.
- **Modality.** Running interferes with leg strength and size far more than cycling, because the eccentric pounding of footstrike competes directly with leg growth and recovery.

When lower-body development is the priority, push conditioning toward low-impact options like the bike or the ski erg, and save running volume for blocks where the legs are not the headline.

- **Dose.** Building a quality is expensive. Holding it is cheap. That gap is the whole basis of concurrent training. It lets you give one quality the lead while the others tick over on a small maintenance dose.

### **Interference, what it really is**

The loudest claim in hybrid training is that cardio kills your gains. Mostly it does not. The molecular story is real but small and very dependent on context, and it is not a hard wall. Almost everything people blame on interference is actually two things you can manage: fatigue and tension.

### **Fatigue, and where it really comes from**

Energy is finite. Total fatigue is paid out of three accounts, and the body cannot tell them apart: training, life, and recovery. Most of what looks like interference is just too much total cost arriving at the same session. The training account is the one everyone counts. The other two decide whether the training even works.

Compare two athletes with the same program. The first is a full-time athlete. They train, eat, nap, and train again. Their job is to recover. Their life account is nearly empty of stress, their recovery account is full, so almost all of their capacity goes into training, and they can handle a big dose. The second person has a demanding full-time job, a commute, a family, deadlines, and broken sleep. They want to train like the first athlete, and on paper they run the same program. But their life account is already drained before they walk into the gym. A poor night of sleep, a stressful week, skipped meals, low water, all of it comes out of the same finite pool that training draws from. Hand them the full-time athlete's program and it will not just feel harder, it will actively break them down, because there is nothing left in the other two accounts to pay for it.

This is one of the most important practical points in the book. You program the person in front of you, not the athlete they wish they were. The amateur with a stressful life often needs less training volume, not more, because their non-training stress is already high. The skill is to read all three accounts and dose the training to what is actually left after life and recovery have taken their share. Push someone whose life account is empty and you are not building them, you are burying them.

### **Tension, the other half**

The same tension that builds muscle also squeezes off the muscle's own blood supply and adds to fatigue. High tension in the right place, at the right time, is exactly what you want. High tension everywhere, all week, with no plan for where it lands, is how concurrent training collapses. Think of it as a budget. A maximal lift, a heavy tempo eccentric, and a hard hilly run all spend from the same tension account, and if they all land in the same forty-eight hours, the body cannot recover any of them well. Spread high-tension work out. Keep the low-tension days genuinely low. Do not let a supposedly easy day quietly turn into another high-tension day, because that is usually where the wheel comes off.

Put those two together and concurrent training is very workable. Separate your hard sessions in time. Respect modality so running does not eat your leg day. Manage the total load across all three fatigue accounts, not just the training one. Watch tension and fatigue

above everything else, because they are the two variables that decide whether your training adds up or falls apart.

## The foundation, fuel, sleep, hydration

None of the programming matters if the foundation under it is missing. The most common failure in hybrid athletes is not their program. It is that they are under-fuelled, under-slept, and under-recovered, asking two demanding systems to adapt on too little of everything. When the inputs are short, the body triages, and survival always beats adaptation.

- **Nutrition.** Match intake to intent. Two demanding systems on too few calories means the body protects itself and drops adaptation. Low energy availability quietly caps strength, blunts conditioning, and erodes bone and hormone health. Stop trying to strip fat and build performance in the same breath, pick one as the headline, and eat for it. Get enough protein to repair the work you are doing, and enough carbohydrate to fuel the hard sessions, especially around the priority work.
- **Sleep.** Sleep is where adaptation actually happens, and it is the cheapest performance lever there is. Short or broken sleep drains the recovery account directly, raises perceived effort, and blunts both strength and aerobic gains. For the amateur with a stressful job, fixing sleep is often a bigger win than any change to the training. Treat it like a session: protect it, plan for it, and judge a hard week partly by whether sleep held up.
- **Hydration.** Even mild dehydration drops output, raises heart rate and perceived effort, and makes a normal session feel like a hard one, which then costs you more fatigue for the same work. Drink to thirst across the day, take in fluid and some electrolytes around long or sweaty sessions, and do not let a hydration problem masquerade as a fitness problem.

These three are not separate from the program. They are part of it. We come back to them as things you prescribe on purpose, with the same intent as a hard interval, later in Running It.

## PART III

# The Programming Engine

*How you progress a movement, fill a domain, build a week, and shape a program*

Part II explained why. This part is how you actually run it. How you advance a movement and walk it back when it breaks, what fills each domain, how the week is built, and how a program comes together over time.

## Module 1, the movement progression engine

This is the logic underneath choosing exercises: how you advance a movement, and how you regress it when execution falls apart. It is a troubleshooting map, not a fixed ladder. You start a movement with every bit of help switched on, then progress by taking that help away, one piece at a time.

### ON SOURCES

These ten principles of progression come from Pat Davidson. I use them here because they are the simplest and most honest progression model I have found, and they fit everyone and every modality. They apply just as well to strength, to skill, and to conditioning. Think of the whole thing as a skill spectrum: every quality can be made easier or harder along the same set of dials.

### The ten principles, with movement examples

- **1. Start static.** Master the position before it moves. Hold a split-squat position before you do a walking lunge.
- **2. Start sagittal.** Front-to-back first, then side-to-side, then rotation, then everything at once. A forward lunge before a lateral lunge before a rotational one.
- **3. Start two limbs, even load.** Before staggered or single-limb. A two-leg squat before a split squat before a pistol.
- **4. Make gravity easy first.** Supported or horizontal before standing and loaded. A hip thrust on the floor before a standing single-leg deadlift.
- **5. Only train the range you can control.** Squat to the depth you own, then expand it on purpose. Do not chase a depth your position cannot hold.
- **6. Start with short levers.** Keep loads and limbs close before lengthening them. A goblet squat with the weight at the chest before a back squat, a bent-knee leg raise before a straight-leg one.
- **7. Add a small wrong-way cue.** A light band pulling the knee the wrong way in a split squat so the body learns to correct itself.
- **8. Give the athlete things to feel.** A wall, the floor, a bench, a band to orient against. A wall-supported press before a free-standing one.
- **9. Limit the freedom.** Take away the wrong options so only the right movement is available. A landmine press, which fixes the bar path, before a free overhead press.
- **10. Add load last.** Load is the final dial you turn up, not the first. Earn the pattern, then load it.



## Progress, regress, and always scale

- **Progress** by removing help or adding demand: static to moving, sagittal to rotational, two limbs to one, short lever to long, supported to free, fewer cues and constraints, and only then more load, speed, or duration.
- **Regress** when form breaks under fatigue or load. Do not throw the exercise out, walk back up the list. Add a constraint back, shorten the lever, return to sagittal, drop the load. Fix the pattern where it holds, then climb again.

Said plainly: not scaling in training is a mistake. Scaling is not giving up, it is the tool. It keeps an athlete training the right pattern at the right quality instead of practising failure. The right to load, lengthen, and speed up is earned rep by rep.

## Know when to break the rules

This is a guide, and a guide can be broken. The point is knowing when. You would never break these rules with a beginner, where the rules are doing exactly the job they exist for. But an athlete with a long training history, high skill, and good recoverability will sometimes do well with them broken on purpose. A CrossFit athlete heading into a competition needs to practise high load with high skill at the same time, because that is what the sport will demand of them. That is specific work, and it is the right call near competition. It is the wrong call year-round, where it adds risk for no adaptive reason. So the rule is: follow the progression to build, break it deliberately to prepare for a known specific demand, and only with an athlete who has earned the right.

## Where it plugs into True Hybrid

The movement progression runs alongside the contraction line and the domains. You do not earn a fast, rotational, single-leg expression while your base is still slow, supported, and two-legged.

| Movement stage                                                                                                                                                                                                                                                                                                                    | Contraction            | Domain              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------------------|
| <b>Static, sagittal, two-leg, short lever, lots of help, light load</b>                                                                                                                                                                                                                                                           | Concentric / isometric | Build, early Endure |
| <b>Controlled tempo, side-to-side, staggered, longer levers, moderate load</b>                                                                                                                                                                                                                                                    | Tempo eccentric        | Endure              |
| <b>Moving, rotational, single-leg, free, fast, reactive, heavy load</b>                                                                                                                                                                                                                                                           | Reactive / SSC         | Express             |
| <b>APPLY IT</b><br>Pick one movement you are stuck on. Find where it sits on the progression, then take one step back, add a constraint, shorten the lever, or drop the load, and own it cleanly for a couple of weeks before you try to progress again. Most stalls are not a strength problem, they are a position you skipped. |                        |                     |

## Module 2, the method library

These are the conditioning methods, sorted onto the spine. Two ideas organise them. First, the domain: each method lives in Build, Endure, or Express. Second, the mode: a method is either cyclical or mixed.



- **Cyclical** means one repeatable action you can pace and measure precisely, like a bike, a row, a ski erg, or slow air squats with built-in rest. Low skill demand, easy to control, easy to read.
- **Mixed** means several movements blended together, often loaded, like a CrossFit-style piece. Higher tension, higher skill, harder to pace, harder to measure, but closer to what a mixed-modal sport actually asks for.

The two modes map onto the tension and skill line. Build is best done cyclical and repeatable, because you want clean, low-cost volume. Endure works both ways: cyclical when you want to measure the engine, mixed when you want to add load and skill at the edge. Express works both ways too, heavier and more skilled in both. Match the mode to what you are trying to train, not to what looks impressive.

One rule governs every method, and it is driven by data: run a method until the numbers stop improving, then change to attack the next limiter. Do not change for variety. Repeat, plateau, move on.

### The method library at a glance, by shade

Each method carries its shade, so the library and the shade table in Part II speak one language. This only pins each method to where it actually sits on the line; nothing about how you run them changes.

| Method                       | Domain  | Shade      | Mode             |
|------------------------------|---------|------------|------------------|
| <b>Aerobic Base</b>          | Build   | Low → high | Cyclical / mixed |
| <b>Tension Base</b>          | Build   | Low → high | Cyclical         |
| <b>Recoverable Intervals</b> | Endure  | Low        | Cyclical         |
| <b>Carry and Brace</b>       | Endure  | Low → high | Mixed            |
| <b>Edge Intervals</b>        | Endure  | High       | Cyclical         |
| <b>Grind</b>                 | Endure  | High       | Cyclical         |
| <b>Loaded Mixed</b>          | Endure  | High       | Mixed            |
| <b>Lactic Pieces</b>         | Endure  | High (top) | Mixed            |
| <b>Peak Intervals</b>        | Express | Low        | Cyclical         |
| <b>Heavy Mixed</b>           | Express | High       | Mixed            |
| <b>Power Repeats</b>         | Z7      | —          | Cyclical         |
| <b>Reactive Repeats</b>      | Z7      | —          | Mixed            |

Two placements are worth a word, because they are the ones people get wrong. **Power Repeats and Reactive Repeats are Z7, not Express high**, short, violent, fully recovered, and cheap to recover from, the opposite profile to the lactic Express work. They sit above the six metabolic shades, not at the top of them. And **Lactic Pieces straddle the Endure–Express seam**: they live at the top of Endure but tip into Express low as the duration shortens and the rest lengthens, which is exactly why the card leans on long rest. Do not let one drift into the other by accident.

**SPRING PREP IS NOT ON THE METABOLIC LINE**

Spring Prep is a real Build-adjacent method, but it deliberately does not appear on the shade table above, because the shades are a metabolic scale, defined by lactate and oxygen, and Spring Prep is not metabolic work. It is neuromuscular and tissue work: laying down elastic, durable tendon and teaching quiet, springy contacts at minimal fatigue. Its whole point is to produce almost no metabolic stimulus, so calling it Build low would be true on cost but false on adaptation. It belongs to the connective-tissue loading ladder and the contraction line in Part II, isometric, then heavy slow, then reactive, as the first rung of reactive tissue prep. Program it when you would program easy work, but read its adaptation off the tissue ladder, not the engine. This is the same reason Power Repeats and Reactive Repeats sit at Z7 rather than inside the metabolic shades: they are as much neuromuscular as metabolic.

## Build, base the engine

**Aerobic Base — cyclical or mixed.** *For:* build delivery, a big, efficient engine at low cost, plus breathing, pacing, and durability. *Example:* Bike 75 min @ RPE 4 (conversation pace, around 65% max HR); or a 50 min EMOM @ RPE 4, each minute alternating 30 sec of a light carry or hold with 30 sec of easy erg. *Feel:* easy all day, unbroken, full conversation throughout. If you are breathing hard, slow down.

**Tension Base — cyclical.** *For:* slow-twitch and local muscular endurance, the concentric resistance base for the engine. *Example:* tempo goblet squat or leg press, 2s down / 2s up at 40–50% 1RM, 8–10 reps x 4–6 sets, rest 45–60 sec @ RPE 4–5, RIR 4+; or 10 x (40 sec air squats + 20 sec rest), smooth and controlled. *Feel:* working, but never near failure. A steady burn, never a grind. Plenty left every set.

**Spring Prep — tissue, not metabolic.** *For:* lay down elastic, durable tissue, soft landings and bounce at minimal fatigue. *Example:* low pogo hops and short box step-downs @ RPE 4, 8–15 quiet, springy contacts x 4–6 sets, 10–30 sec rest. *Feel:* bouncy and easy. Quality of contact over quantity. Stop if landings get loud. No soreness next day. (See the note above: this is the entry rung of the connective-tissue ladder, not a metabolic shade.)

## Endure, hold output at the edge

**Recoverable Intervals — cyclical (Endure low).** *For:* heart-rate recoverability and pacing, the bridge from easy work into intensity. *Example:* Bike 10 sec hard / 60 sec easy @ RPE 6 for 20–30 min; or Row 90 sec @ RPE 6 / 90 sec easy x10. *Feel:* brief surges, then settle. The skill is how fast your breathing comes back down between efforts.

**Edge Intervals — cyclical (Endure high).** *For:* sustained effort right at the edge of what you can hold, threshold work. *Example:* 2 x 18 min Bike @ RPE 7 (hard but talkable, around 85% max HR), rest 5 min; or 3 rounds of 5 min Row / 5 min Bike / 5 min Run @ RPE 7, rest 5 min. *Feel:* working and uncomfortable, held on purpose. You could sustain it, but you would not want to.

**Grind — cyclical (Endure high).** *For:* fast-twitch and local endurance under heavy, continuous output. *Example:* 5–20 min continuous @ RPE 7–8, high-resistance bike at low cadence or a heavy ski, unbroken, holding the output steady as it starts to bite. *Feel:* heavy and relentless. Legs and lungs both complain. Nowhere to hide, no coasting.

**Loaded Mixed — mixed (Endure high).** *For:* tension and skill at the edge, a controlled mixed-modal piece. *Example:* 18 min AMRAP @ RPE 7–8: 18 cal Ski, 18 walking lunges, 18 alt DB

snatch, 30 double unders. *Feel*: strong and holdable. Pace round one like you have to do all of them, because you do.

**Carry and Brace — mixed (Endure low → high).** *For*: trunk, grip, and posture, the bracing entry into Endure. *Example*: 5 rounds @ RPE 7, rest 60–90 sec: 40 m heavy farmer carry, 20 m front-rack carry, 30 sec sandbag bear-hug hold. *Feel*: grip and midline give out before the lungs do. Stay tall, keep breathing, do not round over.

**Lactic Pieces — mixed (top of Endure, into Express low).** *For*: output under heavy metabolic stress, the top of Endure. *Example*: 30–60 sec all-body efforts @ RPE 8, rest 2–5 min (e.g. 12 thrusters + 12 burpees, hard and unbroken), x4–6 sets, long rest so each set stays strong. *Feel*: a deep burn that builds across each effort. The long rest is the method, not a luxury.

### Express, produce maximal output

**Power Repeats — cyclical (Z7).** *For*: pure short-burst power, the very top of the line, quality over quantity. *Example*: 6 x 15 sec heavy sled push, max turnover, rest 3–4 min; or 6–8 x 10 sec bike sprint, all-out, rest 2–3 min. *Feel*: short and violent, then full recovery. Stop the day speed drops. Heart rate is not the guide here.

**Peak Intervals — cyclical (Express low).** *For*: drive the heart to its ceiling, mechanics and breathing under real fatigue. *Example*: 4 x 1 km Row @ RPE 9, rest 3 min; or 12–15 x 60 sec Bike @ RPE 9–10, rest 60 sec. *Feel*: near maximal. You cannot talk. Glad when the rest comes, ready again when it ends.

**Reactive Repeats — mixed (Z7).** *For*: repeatable explosive power, spending the elastic base you built in Spring Prep. *Example*: 5–20 sets @ RPE 9, 60 sec rest: 3–5 broad jumps or box jumps, maximal intent. Cut the session the moment height or distance drops off. *Feel*: explosive and crisp every rep. The instant it slows, you are done, even with sets left on paper.

**Heavy Mixed — mixed (Express high).** *For*: express force and skill together, the way a competition demands. *Example*: 5 min AMRAP @ RPE 9–10: 10 wall balls, 10 burpees, 10 cal Ski, rest 10 min x3 sets; or 21–15–9 thruster + pull-up for time, rest 12–15 min, x3 (advanced only). *Feel*: all-out with full recovery. Each round should be fast, not a survival shuffle.

Notice the bookend. Spring Prep at the bottom of Build lays down the elastic, durable tissue. Reactive Repeats at the top of Express spends it. Same quality, opposite ends of the line, and, tellingly, both live off the metabolic shade scale, on the tissue and neuromuscular track instead.

### What each domain costs, and how often you can train it

The further right you go, the more a session costs and the longer you need before you can repeat that kind of work well. This table is one of the most practical things in the book, because it tells you how to space your hard work. It also gives you the RPE, heart-rate, and talk-test ranges that let you hit a domain without any fancy testing.

| Domain                   | RPE | % max HR | Talk test                          | Recovery before you repeat it                                             |
|--------------------------|-----|----------|------------------------------------|---------------------------------------------------------------------------|
| <b>Build (easy)</b>      | 3–5 | 50–75%   | Full conversation                  | Very low cost. Train daily, twice a day, or back to back with no problem. |
| <b>Endure (moderate)</b> | 6–8 | 75–90%   | Short sentences, then single words | Moderate cost. Most Endure work repeats inside 24 hours,                  |

| Domain                 | RPE  | % max HR | Talk test   | Recovery before you repeat it                                                                                                       |
|------------------------|------|----------|-------------|-------------------------------------------------------------------------------------------------------------------------------------|
|                        |      |          |             | but not smashed daily. The near-threshold end needs about 48 hours.                                                                 |
| <b>Express (hard)</b>  | 9-10 | 90-100%  | Cannot talk | High cost. Needs roughly 72 hours before the next true Express session. The most demanding work you do.                             |
| <b>Z7 (max, short)</b> | Max  | n/a      | n/a         | Short maximal efforts with full rest. Low recovery cost despite the intensity, because the bouts are tiny and the rest is complete. |

Read it the way a coach uses it. You can stack Build work freely, which is why easy aerobic volume is the safest way to grow the engine. You space Express work across the week because it needs three days to recover. And you never let a hard mixed piece masquerade as an easy day, because the recovery clock does not care what you called it.

### Combining methods

Most of the time you run one method and progress it. Sometimes you deliberately bridge two, to train the handoff between domains. The most useful bridge teaches the body to express right after it has been working aerobically, which is exactly what mixed sport asks for.

**Engine Builder / cross-domain.** 4 sets: 3 min Bike building RPE 4 up to RPE 8 across the interval, straight into 15 cal Ski, 20 wall balls, 10 pull-ups, 400 m Run @ RPE 8, rest 4-6 min. *Feel:* no rest between the bike and the burst. That seam, expressing on tired aerobic legs, is the whole point.

### Audit your week

Write every conditioning piece from a normal week into the blank rows, then fill in its domain, its mode, the RPE you actually hit, and its recovery cost. Two examples are filled in to show the idea.

| Your piece                           | Domain | Mode     | RPE | Recovery cost |
|--------------------------------------|--------|----------|-----|---------------|
| <b>Sat mixed grind (long metcon)</b> | Endure | Mixed    | 7-8 | 24-48 h       |
| <b>Tue 60 min easy bike</b>          | Build  | Cyclical | 4   | Low, daily    |
|                                      |        |          |     |               |
|                                      |        |          |     |               |
|                                      |        |          |     |               |
|                                      |        |          |     |               |

#### APPLY IT

Once it is filled in, look at the spread. Most people find they are crowded into one box, usually mixed Endure pieces, because those are fun and feel productive, while their cyclical Build base and their true Express work are missing. A balanced week has work in

| Your piece                                                                                                               | Domain | Mode | RPE | Recovery cost |
|--------------------------------------------------------------------------------------------------------------------------|--------|------|-----|---------------|
| more than one box, with the most volume sitting in the domain that is actually your limiter, not the one you enjoy most. |        |      |     |               |

## Module 3, programming strength and hypertrophy

Part II covered the principles, that tension drives growth, that recruitment comes before volume, that strength is more than size. This module is how you actually program it, and how it blends with the conditioning instead of fighting it. Strength and size are not the same job and they are not trained the same way, so separate them in your head first.

- **Strength** is about producing force. You train it with heavy loads, low reps, and full effort on each rep, leaving a rep or two in reserve to keep the quality high. The nervous system is doing most of the work.
- **Hypertrophy** is about building tissue. You train it with moderate loads taken close to failure, more reps, and more total volume, keeping the muscle under tension. The muscle is doing most of the work.

### Tiers of exercise

Every session sorts its lifts into three tiers, and the tiers get different doses.

| Tier             | What it is                                                                                                                | Typical dose                                                           |
|------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Primary</b>   | The big compound that drives the session. Squat, deadlift, press, pull, or an Olympic variation.                          | Heavy. 3–6 sets of 1–5 reps for strength, 1–2 reps in reserve.         |
| <b>Secondary</b> | A supporting compound that builds the primary and adds quality volume. Front squat, RDL, incline press, weighted pull-up. | Moderate. 3–4 sets of 5–10 reps, near but not at failure.              |
| <b>Accessory</b> | Isolation and health work. The muscles, tendons, and positions that support the big lifts and keep the athlete intact.    | Higher reps. 2–4 sets of 8–20 reps, taken close to failure for growth. |

The hypertrophy dose from Part II lives mostly in the secondary and accessory tiers: roughly eight to fourteen hard sets per muscle in a session, more across the week, split by frequency. Anywhere from thirty to ninety percent of your max can build muscle if it is taken close enough to failure, so you have room to grow tissue with heavy secondary work and with lighter, higher-rep accessory work. The slow tempo eccentrics belong here too, on the secondary and accessory lifts for the muscles you most want to grow, placed where they will not wreck the next priority session.

### How strength rides the same priority logic

Strength obeys the same rule as everything else: bias the limitation, maintain the rest. When strength is the headline, it leads. You put real heavy primary work in two to three times a week, you progress the load, and you let conditioning sit on a small maintenance dose. When strength is the support, the asymmetry saves you. Holding strength is cheap. A couple of heavy primary sets once or twice a week, kept fresh and well short of failure, will hold most of an athlete's strength while the conditioning leads. You do not need to chase a

new max in a conditioning block. You need to remind the system how to produce force so it is still there when you rotate the lead back.

### **Hypertrophy when the legs are the priority**

Hypertrophy is where modality discipline bites hardest. The eccentric pounding of running competes directly with leg growth and recovery, so when you are trying to add leg size, you keep the conditioning low impact, the bike, the ski, the sled, and you save running for blocks where the legs are not the headline. Stack a hard hilly run and a heavy tempo squat day in the same forty-eight hours and neither adapts. Separate them, and both can.

### **Sequencing strength and conditioning in the same block**

Across a block, strength and conditioning take turns leading while the other maintains. In a conditioning-led Build block, conditioning carries the volume and strength holds on two heavy sets a couple of times a week. In a strength-led block, the heavy work carries the volume and conditioning drops to easy Build aerobic to keep the engine without adding fatigue. You never run both at full volume at once, because that is the surest way to stall both. Rotate the lead, hold the rest, retest, and rotate again.

#### **APPLY IT**

Take your current strength work and sort every exercise into primary, secondary, or accessory. Then ask one question: is strength my headline or my support this block? If it is the support, you are almost certainly doing too much of it. Cut it back to a couple of heavy primary sets to maintain, and put the freed-up recovery into your actual limiter.

## **Module 4, session structure and the warm-up**

Every session runs the same shape, and the order is not arbitrary. The highest-quality, highest-skill, most neural work goes first while you are fresh, and the fatiguing work follows. You ramp from low intensity to high across the session, what we call running from level one to level three, so the body is ready for the hard part by the time it arrives.

The order: warm-up, then your priority quality, then your supporting work, then the cooldown. If power or maximal strength is the priority, it leads, because it needs you fresh and it falls apart under fatigue. If the engine is the priority, the conditioning can lead. Either way, the fast and skilled work comes before the deeply fatiguing work, never after it.

### **The warm-up, three jobs**

A warm-up is not ten minutes of easy cardio and a few stretches. It has three jobs, done in order, and each one sets up the next.

- **Raise the temperature.** A few minutes of easy aerobic work to get warm, get blood moving, and lift the heart rate gently. Build domain, easy.
- **Get into position.** The mobility, breathing, and position work that puts the athlete into the structure the session needs. This is where the structure work from Part II lives. Get the ribs stacked over the pelvis, get a full breath, own the positions you are about to load. You cannot recruit what you cannot get into position, so this is not optional, it is the warm-up doing its real job.
- **Rehearse and ramp.** Rehearse the actual patterns of the session and build toward the working load or speed. Light, then moderate, then close to working. For power and lifting



this is your ramp-up sets. For conditioning it is a progressive build into the first hard effort. By the end of the warm-up the body should be ready to express, not surprised by it.

### The cooldown, where recovery starts

The cooldown is the first deposit into the recovery account, and most people skip it. A few minutes of easy Build work to bring the heart rate down, then slow breathing to shift the nervous system from stress toward recovery. This is the same breathing that trains the mental regulate layer, so the cooldown is doing double duty: it starts physical recovery and it trains down-regulation on command. Two or three minutes of slow breathing at the end of a hard session is one of the highest return habits an athlete can build.

### Training position and breathing

Because the position work gets mentioned so often, here is what it actually is, so it is not a black box. The goal is simple to state: own a full, easy breath while stacked in good position, ribs down over the pelvis, so that you can get into and breathe inside the shapes you are about to load. Most athletes cannot, and it quietly caps everything above it.

The core drills are unglamorous and they work. A back-supported 90/90 breath, hips and knees bent over a block, where you fully exhale and feel the front ribs draw down before you inhale low and wide. A dead bug with a long exhale, holding the stack while a limb moves. A supported wall-press breath to feel the back of the ribs expand. Loaded carries done with a long exhale and the ribs down, which trains the same stack under load. None of these are mobility for its own sake. Each one is teaching the athlete to find and keep a position and to breathe inside it.

How to dose it depends on whether structure is the limiter. For everyone, a few minutes of this work belongs in the position block of every warm-up, so the athlete trains in good position rather than reinforcing a bad one. When structure is the actual limiter, the powerlifter stuck in extension who wants to run, the rib-flared athlete who cannot breathe under load, you give it its own focused block three or four times a week and you bias toward the opposite of their default. Extended, braced athletes need exhale-led, ribs-down, slightly flexed work. Loose, flexed athletes need bracing and stiffness they cannot currently produce. Either way it takes months, it runs alongside the training rather than before it, and you measure progress by whether better positions become available under real load.

#### APPLY IT

Audit your own warm-up. If it is just easy cardio, you are missing two of its three jobs. Add a position and breathing block in the middle, get stacked and get a full breath, and a proper ramp into the first hard effort. Then add ninety seconds of slow breathing to the end of your next hard session and notice how much faster you settle.

## Module 5, the weekly engine

Three day types carry the week, each with its own stress budget and job.

| Day type    | Stress | RPE  | Duration  | Role                                                    |
|-------------|--------|------|-----------|---------------------------------------------------------|
| Development | High   | 9-10 | 60-90 min | The expensive day. One or two conditioning methods plus |



| Day type            | Stress   | RPE       | Duration  | Role                                                                                                    |
|---------------------|----------|-----------|-----------|---------------------------------------------------------------------------------------------------------|
|                     |          |           |           | possible max-effort lifting. This is what drives adaptation.                                            |
| <b>Stimulation</b>  | Moderate | 7-8       | 45-75 min | The priming day. Moderate volume and intensity plus one or two core lifts. Sets up the development day. |
| <b>Recovery Day</b> | Low      | 6 or less | 35-45 min | Regeneration. Mostly low intensity, stimulating the body without heavy eccentric load.                  |

A recovery day does not have to be nothing. That is the mistake people make with it, either smashing themselves on a day that should be easy, or doing literally nothing and losing the chance to actively recover. A recovery day can include easy aerobic work in Build, mobility and position work, light technical practice, breathing and down-regulation, and even heavy but very low-volume nervous-system work like a few Olympic-lift or deadlift singles at 85-90 percent, which stimulate the system without the eccentric cost that needs recovering from. The aim is to leave the day feeling better than you started, not worse.

Week shape: two or three development days, one or two stimulation days, one or two recovery days, one or two off. Stimulation comes before development, you prime then push. The session itself always runs the same order: warm-up, conditioning, strength, cooldown.

### Mapped to the domains and the layers

- **Development days** carry the Express and upper-Endure work, plus max-effort lifting, because that is where the stress budget is highest. This is also where the mental express work trains: composure, decision-making, and resetting after a bad rep, which only train when fatigue is real.
- **Stimulation days** carry Endure work with core lifts at moderate load. This is the build layer for the mind too: attention and discipline fit the moderate, repeatable day.
- **Recovery days** are the Build domain by design, easy aerobic and low-tension work. This is the home for the mental regulate work and the spiritual anchor work. Train down-regulation on the days you are already down-regulated.

### Sample week, six training days, the engine leads

This is a full hybrid week, and it shows the core principle in action: one quality takes the lead, and everything else is held at a minimum effective dose. It still blends lifting, size work, and conditioning, but it does not chase them all at once. Here the limiter being chased is the engine, so the conditioning carries the volume and progresses across the block, and the two development days earn their name from their conditioning. Strength and size are not the bias, so they are held cheaply, heavy but low-volume lifting to keep force, and light accessory to keep size at sufficiency. The legs are still trained twice, but to hold them, not build them, and the two days are kept different in cost so they do not compound.

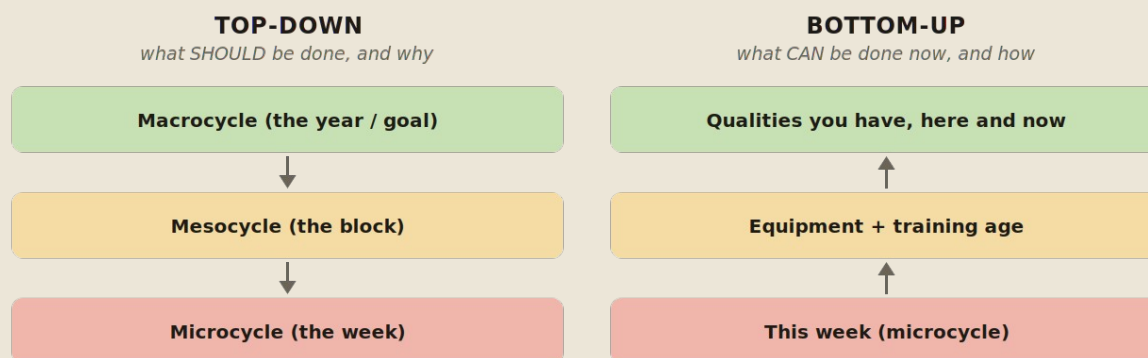
| Day        | Type | Strength and hypertrophy                                                                               | Conditioning                        | Layer           |
|------------|------|--------------------------------------------------------------------------------------------------------|-------------------------------------|-----------------|
| <b>Mon</b> | Dev  | Power first: cleans / box jumps. Then back squat 4x4 heavy and RDL 3x8, held to maintain, not to grow. | Endure, Edge Intervals on the bike. | Mental: express |

| Day        | Type                  | Strength and hypertrophy                                                                                                | Conditioning                                                      | Layer                    |
|------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------|
| <b>Tue</b> | Stim                  | Upper, held: bench or push press 4x3, weighted pull-up 4x5, DB row 3x10.                                                | Short Express intervals (high but brief, low fatigue cost).       | Mental: build            |
| <b>Wed</b> | Recovery              | Low-CNS accessory only: arms, upper back, calves 3x12-15. Optional heavy trap-bar singles to feel force.                | Build, 40 min easy bike. Mobility and breathing.                  | Regulate + anchor        |
| <b>Thu</b> | Dev                   | Concentric legs to hold, not build: front squat 4x6 controlled and sled push, no tempo eccentrics. Posterior accessory. | Endure mixed, harder: Loaded Mixed or Lactic Pieces (low impact). | Mental: express          |
| <b>Fri</b> | Stim                  | Trap-bar deadlift 4x3 to hold strength concentrically. Hamstring + trunk accessory.                                     | Endure, moderate. Recoverable or Edge Intervals.                  | Mental: build            |
| <b>Sat</b> | Recovery / long Build | Optional easy sled drags and loaded carries, no real load.                                                              | Build, 60-90 min easy aerobic (low impact).                       | Spiritual: long practice |
| <b>Sun</b> | Off                   | -                                                                                                                       | Rest, reflection.                                                 | -                        |

Read the principle in it. The engine is the lead, so the conditioning is where the volume sits and where the progression happens across the block, and the two development days, Monday and Thursday, are developmental because of their total stress, not because of the lifting. All the lifting is maintenance: heavy enough and often enough to hold strength and size, never enough to chase them, which is exactly the minimum effective dose the model runs its support qualities on. The two leg days are kept apart and kept different in cost, Monday heavy and neural, Thursday concentric and conditioning-led with the eccentrics pulled, so holding the legs twice a week does not eat the recovery the engine needs. Tuesday is a stimulation day because it is upper-body only with short Express intervals that cost little. Friday is the lighter pulling day. Recovery days stay genuinely easy, and the conditioning is kept low impact to keep total leg fatigue in check, so nothing competes with the engine, which is the one quality this block is actually trying to move. That is the whole model in a week: bias the limiter, hold everything else, and let no support quality steal from the lead.

## Module 6, building the program

A week becomes a program when you decide how it changes over time. There are two honest ways to build, and good programming uses both. They answer different questions.



*Two directions of planning. Top-down starts from the goal and works down to the week. Bottom-up starts from what the athlete can do right now and works up. You need both.*

### Top-down, what should be done

Top-down planning starts from the destination and works backwards. You picture the whole year or the goal, break it into blocks, and break the blocks into weeks. It asks what should be done, and why. This is where you decide that the athlete needs an aerobic base before a strength peak, or that a competition in twelve weeks means strength leads now and skill leads later. Top-down keeps you pointed at the goal and stops you from chasing whatever feels good this week.

### Bottom-up, what can be done

Bottom-up planning starts from reality. It asks what this athlete can actually do right now, given the qualities they have, the equipment they own, their training age, and how much life and recovery they have left in the tank. It asks what can be done, and how. A perfect top-down plan that ignores a beat-up, time-poor athlete with a home gym is useless. Bottom-up keeps the plan honest and doable.

Good programming holds both at once. Top-down sets the direction. Bottom-up keeps it real. When they disagree, and they will, you bias the limitation the goal demands while staying inside what the athlete can actually recover from this week.

### Proactive and reactive programming

There is a second pairing that sits on top of this, and it is the difference between writing a plan and running one.

- **Proactive programming** is the plan you write in advance. It is your best guess at the right sequence, based on the goal and the assessment. You need one, because without a plan you are just reacting to noise.
- **Reactive programming** is what you change once the athlete starts responding. Readiness drops, a number plateaus, life blows up, sleep falls apart, and you adjust. The plan is a hypothesis, and the athlete's response is the data.

The skill is to be proactive enough to have a real direction and reactive enough to change it when the body tells you to. Lean too proactive and you grind an athlete into the ground following a plan that stopped fitting weeks ago. Lean too reactive and you have no direction at all, just a string of sessions that feel right and add up to nothing. Hold both: a clear proactive plan, steered reactively by how the athlete responds. This is the same idea as bias the limitation, but train the full system, now applied across time.

**APPLY IT**

Write your next month two ways. First top-down: what is the one goal, and what has to lead to get there? Then bottom-up: given your equipment, your training age, and how much stress your life is already carrying, what can you actually recover from each week? Where the two disagree, the honest answer is almost always less volume on the non-priority work and more focus on the limiter. Then leave yourself a rule for reacting: if readiness drops for several days, what do you cut first?

## PART IV

# The Mental Layer

*Command, whether the capacity shows up when it counts*

This is the layer that decides whether the capacity you built in Part II actually shows up under pressure. It runs the same arc as the body, on the same nervous system: regulate, build, express.

## Regulate, condition the nervous system

Start by learning to settle your own physiology on command. There is a named principle underneath this, the psychophysiological principle, described by Green, Green, and Walters back in 1970: every change in your physical state comes with a change in your mental state, and every change in your mental state comes with a change in your physical one. They move together, always, in both directions. This is the concrete version of the shared-nervous-system claim this whole book rests on. It is not a metaphor. Your body and your mind are running on one system, and breathing is the lever you can grab, because it is the one part of that system you can run by hand.

Think of it as a gas and a brake. The stress branch of your nervous system is the gas: it switches you on, raises heart rate and breathing, gets you ready to work or fight. The recovery branch is the brake: it slows you down, settles you, lets you recover and think clearly. The brake is faster to engage than the gas, which is the whole reason this is trainable. You build the brake first, on purpose, so that you can use the gas hard and still come back down. An athlete who cannot down-regulate cannot recover between efforts and cannot think straight under load. And this is the same recovery state that biases Type 1 function in the muscle in Part II, which is why one slow-breathing habit pays off in both the body and the mind.

## The breathing toolkit

You need two directions: a way to settle down and a way to switch on. Most people only ever need to train the first.

- **To settle down**, slow the breath and stretch the exhale. Breathe diaphragmatically at around five to seven breaths a minute, roughly four seconds in and eight seconds out, for five minutes. The long exhale is what trips the brake, so the exhale being longer than the inhale matters more than the exact counts. A 4-7-8 pattern works too, in for four, hold for seven, out for eight, and shorter ratios like 3-6-7 or 2-5-6 are fine if that is long. For an instant settle in the moment, use a double inhale then a long exhale: a full breath in, a second short sip of air at the top, then a slow release. That one drops arousal in a single breath.
- **To switch on**, do the opposite. Longer, more vigorous inhales with short exhales, a few cycles of thirty seconds to a minute, when you need to come up before a hard effort and you are flat.

Dose it like training. Five minutes a day to build the brake, ideally on recovery days and in your cooldowns where you are already settling. One to three minutes at transitions through the day. One to three breaths inside a reset when you need to drop arousal right now. One

warning: do not run the settling version right before you need to be switched on, it works, and it will flatten you.

### **Find your line**

There is a sweet spot of arousal for performance, the old Yerkes and Dodson curve from 1908: too little and you are flat, too much and you fragment. The catch, which Hanin's work on the individual zone of optimal functioning makes clear, is that the sweet spot is not the same for everyone or for every task. Fine, complex, decision-heavy tasks want lower arousal. Raw speed, power, and strength want higher. And some people simply run better hyped while others run better calm. So the skill is three steps: know what the task in front of you demands, know honestly whether you are above or below your line for it, and then breathe up or down to get there. That is what the toolkit is for.

### **Train outside, the oldest regulator**

There is a lever for the nervous system that costs nothing, needs no app, and was the default for every human until about a century ago: go outside and move. Light, air, terrain, and water are inputs the body is built to expect, and modern indoor life quietly removes all of them. Putting your easy work outdoors is one of the cheapest things you can do for the regulate layer, and it happens to pay into the body and the spirit at the same time.

### **The science, light first**

Morning light is the big one. Getting bright light into your eyes early in the day, a habit popularised by Andrew Huberman and built on decades of circadian biology, sets your internal clock for the day. It triggers a healthy, early rise in cortisol, which is exactly what you want in the morning: alertness, focus, steadier mood, and a primed immune system. And it starts a timer, roughly sixteen hours long, that releases melatonin at the right time that night, so morning light is also how you earn good sleep. Outdoor light is ten to a hundred times brighter than indoor light even under cloud, and the eyes, not the skin, are the main route, so a few minutes outside within an hour of waking does more than an hour under office lighting. Do not wear sunglasses for it, and never look straight at the sun.

The rest of the daylight matters too. Regular daytime light supports the systems behind mood and drive, and living in dim indoor light all day is linked with worse mood and worse sleep. Sunlight on skin builds vitamin D. None of this is exotic, it is simply the dose the body evolved to get and no longer does by default.

### **Movement through open space quiets the alarm**

Moving forward through the world under your own power, walking, running, riding, does something specific to the brain. The streaming of the visual field as you move, called optic flow, has been shown to lower activity in the circuits that generate fear and anxiety. Self-generated forward motion turns down the alarm. Wide, panoramic vision, the kind you use outdoors scanning a horizon or a trail ahead, does the same, shifting you out of the narrow, tunnelled gaze of stress into a calmer state. So an easy outdoor run or ride is not only conditioning, it is a moving regulation drill: light in the eyes, optic flow settling the threat system, breathing you can pace. Indoors on a treadmill you get the training and almost none of this.

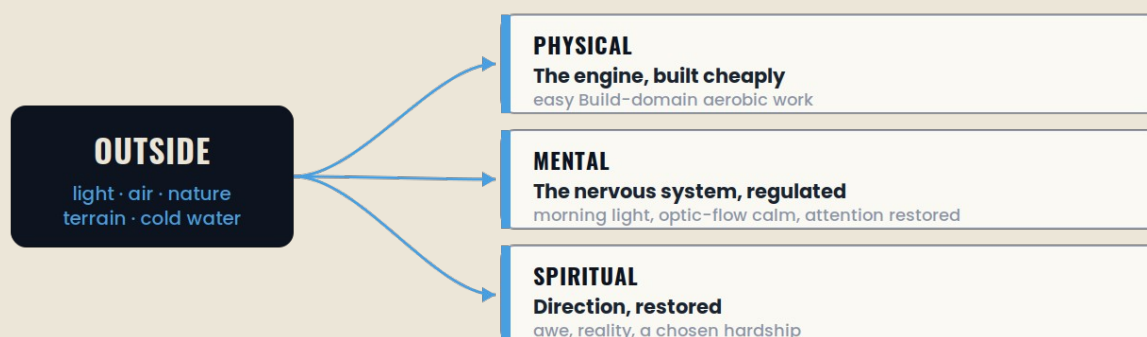
## Air, cold, and green

Time in nature lowers cortisol and blood pressure and restores the kind of directed attention that screens burn through, which is why a walk in a park clears your head in a way a walk down a corridor does not. Colder outdoor air and, best of all, cold water, an open-water swim or a cold dip, drive a large, clean rise in dopamine and adrenaline that lifts mood and focus for hours and builds genuine stress resilience, and it doubles as breath-control practice under real pressure. These are the inputs the regulate layer is built on, delivered for free the moment you step outside.

## The philosophy, get back into the world

There is a reason this feels like more than a health hack, and it is worth saying plainly. Humans were made to live outdoors, engaged with a real and unpredictable world. Modern life is climate-controlled, screen-mediated, and predictable, which is comfortable and quietly deadening. The line of thought Jordan Peterson has pressed hardest is relevant here: voluntarily taking on difficulty, and confronting reality rather than a representation of it, is how a person stays oriented and finds meaning. Weather, terrain, distance, and cold are a small, honest hardship you choose, and choosing them is a form of aiming up. Nature also does something the gym cannot, it sets you against something vastly larger than yourself, and that awe quiets the endless internal chatter and puts you back in proportion, which is the same move the spiritual layer's anchor asks for. Getting outside is not decoration on the training. It is a standing appointment with reality.

## One trip out, three layers trained



*One choice, three returns. Taking your easy aerobic work outside builds the physical engine at low cost, regulates the nervous system through light and motion, and feeds the spirit through awe and voluntary hardship.*

Here is why this belongs in True Hybrid and not in a wellness pamphlet. The easy, low-cost conditioning from the Build domain, the long slow aerobic work, is exactly the training best done outdoors, and hiking, running, riding, and swimming are its natural forms. So a single decision, take the easy work outside, pays into all three layers at once. The body gets its delivery base built cheaply. The mind gets light, optic flow, and nature doing the regulate work for free. The spirit gets awe and a chosen hardship. One nervous system, one calendar, one trip out the door. That is the whole model compressed into a morning hike.

The modalities each bring something. Hiking adds terrain, load, and time on feet at almost no cost. Running banks the eccentric, impact-ready durability the bike never will. Riding lets you pile on aerobic volume with low impact when the legs need protecting. Swimming, especially open water, adds breath control under load and, cold, the biggest regulation hit



of all. Rotate them the way you rotate everything else, biased to what the body needs and what the season allows.

**APPLY IT**

Build the outdoor dose into your week without adding load. First, get outside into daylight within an hour of waking, five to ten minutes on a clear day, longer if it is overcast, no sunglasses, and never staring at the sun. Second, move some of your easy Build and recovery-day aerobic work outdoors, a walk, an easy run, a ride, or a swim, so a real chunk of your weekly easy volume happens under the sky rather than on a machine. Third, make one long, slow, outdoor session a weekly keystone, the kind of single non-negotiable habit the goal ladder is built on. When you can, add cold, an open-water swim or a cold finish, for the dopamine and the breath practice. Pair the easy outdoor work with slow nasal breathing and a wide, panoramic gaze, and you are training the regulate layer and the engine in the same session.

**CAVEAT**

Outdoors is mostly Build-domain and regulate work, an addition to the structured training, not a replacement for it. Respect the obvious safety: never look directly at the sun, manage sun exposure and sunburn, hydrate and account for heat, and progress cold-water exposure gradually and never alone. The point is to reclaim inputs modern life removed, at a dose that helps, not to turn every session into an expedition.

## Build, train attention and mindset

### Where your attention is

The single most useful distinction here comes from Gardner and Moore: attention is either self-focused or task-focused. Self-focused means you are in your own head, judging yourself, stuck on the last rep or the next one, chewing on things you cannot control. Task-focused means you are absorbed in the thing in front of you, in the present, on what you can actually affect. Their point, and it is the whole game, is that peak performance is absorption in the task rather than absorption in the self. Self-focus has its place, after the session, when you review and critique. During the effort it is the enemy, and learning to notice it and drop back into the task is a trainable skill.

### Attention is trainable, and refocusing is the rep

Attention behaves like a muscle. The research on attention training shows that four to eight weeks of consistent daily practice improves focus, the ability to reset after a mistake, emotional steadiness, and the ability to tell what matters from what does not in the moment. The mechanism is the part people get wrong: when you sit to focus and your mind wanders and you drag it back, that drag back is not a failure, it is the repetition. The more times your attention wanders and you return it, the more training you are getting. So a wandering mind in practice is not a problem, it is reps.

The drills are simple. Rest your attention on an external object and return it each time it drifts. Rest it on the breath. Run a slow body scan, noticing sensations without judging them. Open up to broad situational awareness, taking in the whole environment. Then practise shifting deliberately between these. Three to ten minutes a day is enough to start. Then

carry it into training by single-tasking, giving the set in front of you your whole attention and nothing else.

### **Threat or challenge, you can choose the read**

How you read a hard situation changes your body's response to it. Crum's work on stress mindset shows that seeing a stressor as a challenge rather than a threat produces a measurably different and more useful physiology, even though the stressor is identical. A study of Navy SEAL training found that candidates who held a stress-is-enhancing mindset lasted longer and performed better than those who saw stress as purely harmful. Same events, different read, different outcome. This is not pretending something is easy. It is deciding it is a test worth taking rather than a danger to survive, and that decision reaches all the way down into your heart rate and your breathing.

Underneath attention and mindset sit discipline, habits, and values. Discipline and routine are the scaffolding, and you build them in calm so they hold under load. But scaffolding has to be anchored to something, and what it anchors to is your values, what you are actually about. That is where this layer hands off to the next one.

### **Express, perform under pressure**

Regulate and build happen in calm. Express is what you actually run when it counts, and it is a small set of routines you have rehearsed enough that they fire on their own.

### **Recognise, reset, refocus**

This is the core loop, and it is the same shape as the body's express work. Recognise that your attention has gone self-focused or your arousal is off. Reset with a breath or two, taking in your surroundings. Refocus on the next thing you can actually control. The ability to run this loop fast is the entire difference between one bad rep and a bad session. Rehearse it as a daily rep with a simple drill: a few times a day, name three things you can see, three sensations or thoughts you are having, and take three slow breaths, then say where your attention should be right now. Done daily, the recognise-reset-refocus loop gets so quick you stop noticing you are doing it.

### **Control the controllables**

Spending energy on things you cannot control drains you and leaves you flat. Before a hard event, split it in your head into what you can control and what you cannot, and decide to spend nothing on the second list. What you can always control is short: your mindset, your effort, your attention, your breathing, and your next action. That is where the energy goes.

### **See it before you do it**

Visualization is a real rep, not a pep talk. The brain runs a vividly imagined action through close to the same machinery as the real one, which is why imagery research calls it functionally equivalent and why it transfers. To make it work, follow what Vealey and others lay out: make it vivid by using all your senses, not just a picture but the sound, the feel, the effort. Make it controllable by rehearsing it going right, because picturing yourself failing trains failure. Run it from inside your own eyes, at the real speed of the actual effort. Use it three ways: a daily block of ten to twenty minutes on the things you most want to sharpen, a short five to ten minute rehearsal before a key effort, and a review afterward to correct what went wrong and bank the rep.

## Goals that actually drive

There are three kinds of goal, and most people lean on the weakest one. Outcome goals are about the result, winning, placing, hitting a number, and they are largely out of your direct control. Performance goals are about beating your own previous mark. Process goals are about the specific things you do in the effort. Lead with process and performance goals, because those are the ones you can actually move, and let the outcome take care of itself. Then pair them with if-then plans: decide in advance that if a specific obstacle shows up, you will do a specific thing. Pre-deciding the response means an obstacle triggers a plan instead of a scramble. This is the mechanics of a goal. The spiritual layer takes the same goal deeper, down to the person you would have to become to reach it and the one habit that gets you there, in the goal ladder.

## The pre-performance routine

Tie it together into one short, identical sequence you run before every key effort. A breath to set your arousal to your line, a cue word for the one thing that matters, then act. The power is in it being the same every time, so under pressure you drop into a familiar groove instead of a scramble.

## Sustaining, not burning out

Program these the way you program everything else, onto the day types. Regulate work, the settling breath, lives on recovery days and in cooldowns where you are already coming down. Build work, the attention drills and the reframe, fits the moderate stimulation days. Express work, the routines and resets and visualization, only truly trains on the hard development days, because composure under fatigue cannot be rehearsed when you are fresh. One nervous system, one calendar.

## The failure ladder

Push hard without recovering and you climb a ladder you do not want to climb. Kellmann's work on the recovery-stress balance lays out the stages. First is under-recovery: extra soreness, poor sleep, low energy, a short fuse. It is cheap to fix at this stage with simple, deliberate recovery. Let it run and it becomes overtraining: chronic fatigue, a raised resting heart rate, mood changes, performance falling off, and now it needs weeks to months to come back. Let that run and it becomes burnout: exhaustion and the urge to quit the thing entirely. The lesson is to catch it at the bottom of the ladder, because the bottom is cheap and the top is expensive.

## The four states

Schwartz frames it as four states you move between. The performance state is high energy and productive, where you want to be when it counts. The recovery state is low energy and restoring, which is what refills the first one. The survival state is high energy but unproductive, stuck on the gas, anxious, threat-based, spinning. The burnout state is low energy and unproductive, flat and done. The skill is to move deliberately between performance and recovery, because if you never deliberately recover, you drift into survival, and survival is the on-ramp to burnout.

## The daily readiness check

This is the mental-layer twin of the monitoring in Part VII. A brief, honest daily self-rating, how physically ready or fatigued you feel, how mentally ready or foggy, whether you are reading the day as a challenge or a threat, works as an early-warning system. The numbers feed the same back-off decisions as your resting heart rate and your bar speed. Keep it simple. A few honest scores you actually log every day beat a detailed instrument you abandon in a week.

### APPLY IT

Build your routine this week. Pick one settling breath, one attention drill, and one reset, and practice them in calm, away from any pressure. Then use them for real: run the settling breath in your next cooldown, run the reset between sets in your next hard session, and run a short pre-performance routine before your next key effort. You are not adding much time. You are training the six inches that decide whether the rest of it shows up.

### CAVEAT

Skills are trained, not summoned. The mind, like the body, will not produce a skill under pressure that it never built in calm. Train your attention and your breathing when it is easy, so they hold when it is not.

### CAVEAT

Performance is not therapy. Coaching the performing mind is not treating the struggling one. Know the line. When something is mental health rather than mental performance, the move is to refer, not to coach.

### ON SOURCES

The mental layer leans on people who built this field. The psychophysiological principle is Green, Green, and Walters. The split between self-focused and task-focused attention, and the read of mental toughness as acting on your values under pressure, are Gardner and Moore. The work on stress mindset, including the Navy SEAL study, is Crum and her colleagues. The individual optimal-arousal idea is Hanin, sitting on the older Yerkes and Dodson curve. The imagery guidance follows Vealey, the if-then plans follow Oettingen and Gollwitzer, the recovery-stress ladder is Kellmann, the four states are Schwartz, and the readiness check follows Keegan and colleagues. The outdoor-training section leans on the circadian and vision science popularised by Andrew Huberman, which itself draws on the wider fields of circadian biology and visual neuroscience, and its philosophy echoes Jordan Peterson on voluntary responsibility and confronting reality. The way these are organised for performance prep owes a debt to a military cognitive-performance manual by Ben Wellborn, adapted here for the athlete. The science is theirs. The job here is to translate it and put it on the calendar.

## PART V

# The Spiritual Layer

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*Direction, what the whole thing is aimed at*

Capacity and composure still need somewhere to point. This is the layer most people have never trained, and the one that gives the other two their meaning. Same shape: anchor, aim, embody.

## Anchor, stand on something bigger

A person without an anchor drifts. The base of this layer is having something larger than yourself to answer to, the ground everything else is built on. It is the equivalent of regulate. It holds you steady when everything else is moving.

There is a practical handle on this, and it is your values and your why. Frankl and, more recently, Simon Sinek both make the same point: a person who knows why endures things that break a person who does not. People who lose their why quit, even people who are physically more than capable. So this is worth doing on paper, not just in your head. Name your handful of core values, the things you are actually about. Turn each one from a noun into something you do by adding a verb to it, so challenge becomes seek out hard things and discipline becomes show up when you do not feel like it. Then find the why underneath your training by asking why about five times, each answer feeding the next question, until you hit something that does not reduce any further. Write that in one sentence. That sentence is what you reach for when the work stops being fun, and it is the anchor the rest of this layer is tied to.

## Aim, aim up

Point yourself at the highest good you can imagine, and willingly take on the responsibility that comes with it. Meaning over the easy option. Carry the heaviest load you can, not as punishment, but because a load aimed at something gives your effort a direction instead of just motion. This is the heart of the layer: aiming, not just moving.

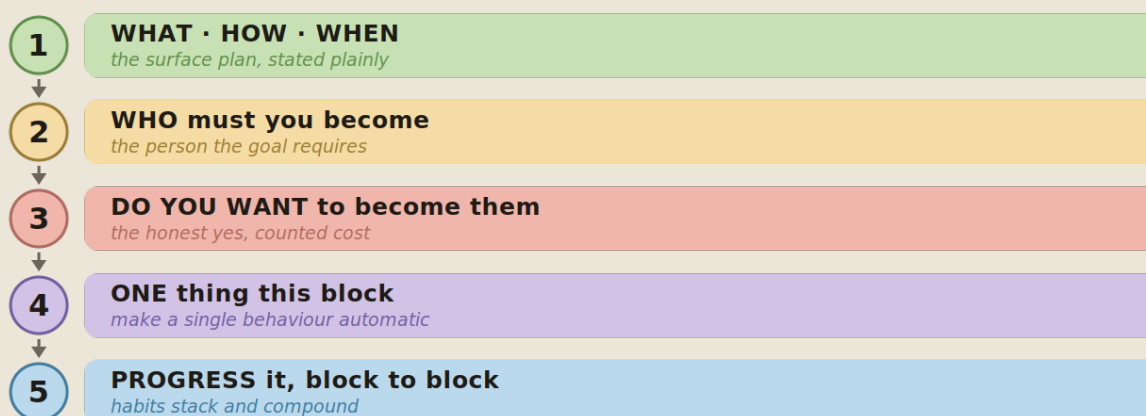
This is the idea Jordan Peterson has put at the centre of his work, and it is worth taking seriously. Life carries a real weight, and a person can meet that weight with resentment, avoidance, and drift, or they can voluntarily pick up a load and aim it at the highest thing they can conceive. Aiming up is what orders the chaos. It sets a direction everything else can be measured against, it turns suffering into something with a point, and it pulls a person toward who they could be rather than letting them settle into who they are. For a man in particular the call is to stop drifting, adopt responsibility on purpose, and orient himself toward an ideal, because a man with nothing above him to serve tends to fall into bitterness or aimlessness, while a man with something above him has a reason to become formidable. The aim is the cure for the drift.

What you aim at is yours, and it shows up in different lives in different ways. The religious man aims up because he is accountable to God and wants to use the gifts he was given as well as he can. The father aims up by working harder and carrying more so his family has a better life than he did. The athlete aims up by chasing a standard just out of reach and

becoming the person who can meet it. Each is the same move: something above the self, freely served, that gives the effort a direction. The point is not which one you pick. The point is that you have one.

## Aim into a plan, the goal ladder

Aiming up is a direction, not a to-do list. To act on it you have to bring it down out of the sky and turn it into something you do on an ordinary Tuesday. Most goal-setting starts and stops at the surface, name the goal, sketch the plan, and then wonders why nothing changes. That surface is real and it is where we begin, but it is only the top of a ladder that goes much deeper. Climb all five rungs and a goal stops being a wish and becomes a person you are building.



*The goal ladder. The top rung is the ordinary plan. The deeper rungs turn it into an identity and then into one repeatable habit that compounds block to block.*

**What, how, and when.** The ordinary plan, and it is the same top-down thinking from the programming engine turned on your own life. What exactly is the goal, stated so plainly you would know for certain whether you hit it. How you will get there, the path and the work it honestly takes. When, a real timeline, not someday. If you cannot write the what, the how, and the when in a sentence each, you do not have a goal yet, you have a mood. Necessary, but this is where most people stop, and it is not enough on its own.

**Who do you have to become.** Here is the rung almost everyone skips, and where this layer earns its place. A real goal is never only a thing you acquire. It is a person you have to become in order to reach it and to hold it. The runner you are not yet, the lifter with habits you do not yet keep, the competitor who stays composed exactly where you currently come apart. Name that person on paper. What do they do that you do not, what do they tolerate that you avoid, what is ordinary for them that is still hard for you. The goal is sitting on the far side of becoming them, and you cannot get the goal without the becoming. This is the old line that the point of a goal is not what you get but who you become on the way, taken literally and made the plan.

**Do you want to become that person.** This is the hard question, and the honest one, and because it is a question about what you value, it belongs to this layer more than any other. Becoming the person your goal requires has a price, in time, in comfort, in other things you will not get to do. Look clearly at who you would have to become and ask, plainly, whether you actually want to be them. Sometimes the answer is no, and that is a gift, not a failure,



because it hands you back a year you were about to spend half-hearted on something you did not truly want. And if the answer is yes, it is a different kind of yes than a wish. It is a commitment made with open eyes, and it is anchored, because it runs straight back to the values and the why at the base of this layer. That yes is what you reach for on the days the work stops being fun.

**The one thing this block.** You do not become that person all at once, and trying to change everything at once is exactly why most attempts collapse in a fortnight. So borrow the core discipline of the whole book and point it at yourself. Bias the limitation. Pick the single behaviour that most moves you toward the person you named, and make it non-negotiable for one block. Not ten habits, one. In the first block it can be as plain as this: you hit every session, no matter what, and everything else is allowed to stay messy. That is the entire job. You are not trying to be optimal, you are trying to make one thing automatic, because a behaviour you have made automatic no longer costs willpower to run.

**Progress it, block to block.** Once that first thing is a habit rather than a fight, something you do without deciding to, it drops to maintenance and you add the next one. This is progressive overload applied to who you are becoming, the same logic the body runs. Block one, you never miss a session. Block two, with that now automatic, you add the daily thing that keeps you moving, dialling in sleep, or fuelling the priority sessions, or a few honest minutes of the mental work. Block three, the next thing. Each habit, once owned, holds cheaply while you build the next, exactly the way a maintained quality holds while you bias a new one. Stack a handful of these across a year and you have not merely worked through a plan, you have quietly become the person the goal always required. And this is the part worth sitting with: every session is a vote for that person. You are not only training, you are casting evidence, over and over, for who you are becoming.

#### UNDER THE HOOD

Why this beats gritting your teeth. Willpower is a poor engine because it runs out, so a plan that depends on wanting it every day will fail on the days you do not. Identity and habit are the durable engines instead. James Clear frames a habit as a vote for the person you are becoming, which is why the who and the one-thing rungs matter more than the plan on top. Charles Duhigg's keystone habits are the reason you pick one behaviour and not ten, because the right single habit drags others along behind it. And the block-to-block stacking is nothing more than this book's own progression logic, own one quality before you load the next, applied to the person instead of the barbell. The science of who you become is not separate from the science of how you train. It is the same science, aimed one level up.

#### APPLY IT

Walk the ladder for one real goal, on paper, now. Write the what, the how, and the when in a sentence each. Then name the person who could reach it, and be specific about what they do that you do not. Ask yourself honestly whether you want to become them, and if the answer is no, pick a better goal. If yes, choose the single thing you will make non-negotiable for the next block, keep it small enough that you cannot fail it, and note what block two might add once this one is automatic. That page is your spiritual layer made concrete.



## Embodiment, live it out

Conviction in action. The point where physical discipline, mental composure, and belief stop being separate practices and become one identity. The integration is the expression.

Gardner and Moore give the cleanest definition of what this looks like under load. Mental toughness, properly understood, is acting toward your values in a purposeful, consistent way even, and especially, when strong emotions are pulling you the other way. Read that carefully, because it is not the absence of fear, doubt, or the urge to quit. Those show up for everyone. Embodiment is moving toward what you are about anyway, with the fear still there. That is why this layer cannot be faked or summoned on the day. It is the values from the anchor and the composure from the mental layer, lived out in the moment it costs something.

### UNDER THE HOOD

Meaning changes how much you can carry, and it does it through the nervous system. A person aimed at something higher can tolerate more, because meaning shifts how they read stress, from threat toward challenge, the same mechanism as the mental layer. Direction changes how much weight someone can carry before it crushes them. That is why this layer is not decoration. It raises the ceiling on the other two.

### CAVEAT

This is an invitation, not a rule, offered as a lens rather than a law. The claim is never which anchor you choose, only that you have one. Anyone who reads this and goes looking for their own grounding, wherever it leads, has taken the point. The invitation stays open and is never forced.

## PART VI

# Integration, True Hybrid

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*Three layers, one nervous system, one calendar*

True Hybrid is not three programs stacked on top of each other. It is three layers that share a nervous system and point the same way.

- **Physical is capacity.** Something to command.
- **Mental is command.** The ability to express that capacity when it counts.
- **Spiritual is direction.** Where it is all aimed.

Train one and neglect the others and you cap the whole system. A strong body with no command chokes. A disciplined mind with no direction drifts. Conviction with no capacity is just talk. The complete picture is the integration: warrior, scholar, and person of conviction as one.

And the integration is practical, not just an idea. The day-type model is where it lives. Recovery days build the regulate and anchor base of all three layers. Development days train physical and mental expression together. The way the program phases over time moves all three through regulate, build, express in step. One nervous system, one direction, one calendar.

***Build the base. Earn the expression. Aim it higher.***

## PART VII

# Running It End to End

*Test, profile, find the limiter, program, monitor, retest*

The whole system runs on one loop: test, program, train, monitor, adjust, retest. This part is the operating manual. How you assess an athlete, find the one thing holding them back, build the program around it, and steer by how they respond.

## 1. Assess

The first rule of testing is the cheapest to say and the most often broken: test what you will program. If you cannot repeat a test, you cannot read change from it, and you are guessing. There are two kinds of testing, and they do different jobs.

| Prescriptive testing                                                                                                                                                 | Descriptive testing                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Reproducible, controlled, cyclical work.</b>                                                                                                                      | Messy, mixed, sport-like work.                                                                                                  |
| <b>Examples: a bike or row Critical Power test, a one-rep-max or rep-max lift, a measured time trial over a set distance, a paced interval set at a known power.</b> | Examples: a benchmark mixed-modal workout, a VO2 or lactate test, a step test, a sport-specific piece scored on time or rounds. |
| <b>Gives you numbers you can write straight into a program. Power targets, loads, paces.</b>                                                                         | Describes the athlete's state and where they break, but it is harder to turn into an exact dose.                                |

Use both, but know which is which. Lead with prescriptive tests when you want numbers to program against. Use descriptive tests to understand the athlete and find where the wheels come off. A benchmark workout is a competition, not a precise test: it is great for describing, poor for prescribing an exact dose, because it blends skill, pacing, and fatigue into one score.

### A menu of tests, and how to use each

Here are the tests worth running, split by type, with what each one gives you and how you actually use it.

| Test                                                   | Type         | What it gives you and how you use it                                                                                                  |
|--------------------------------------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Critical Power test (3 min + 12 min)</b>            | Prescriptive | Your sustainable power and your short reserve. Turns the generic zones into real watts you program conditioning against.              |
| <b>Distance time trial (2k row, 5k run)</b>            | Prescriptive | Your pace at the edge. Sets the exact paces for running and rowing intervals.                                                         |
| <b>Rep-max test (3 to 5 reps) on the main lifts</b>    | Prescriptive | Working loads for strength. Sets the weights and the progression on squat, hinge, press, pull.                                        |
| <b>Step test on the bike (rising watts each stage)</b> | Descriptive  | Where effort, heart rate, and form start to come apart. Shows the intensity at which the athlete breaks, which points at the limiter. |

| Test                                    | Type        | What it gives you and how you use it                                                                                  |
|-----------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Benchmark mixed workout</b>          | Descriptive | How the athlete holds up under real mixed fatigue. Watch where skill and pacing fall apart, not just the final score. |
| <b>Movement screen</b>                  | Descriptive | Structure and position limits. Tells you what to fix before you load a pattern.                                       |
| <b>Resting heart rate and HRV trend</b> | Descriptive | Readiness over time. Used day to day to decide how hard to push, not as a one-off number.                             |

The two types work together. Descriptive testing tells you what the limiter is and lets you watch the athlete break down. Prescriptive testing gives you the numbers to train that limiter tightly. Take a runner who fades late in races. A descriptive step test and a look at their gait under fatigue shows the limiter is structure and breathing. A prescriptive five-kilometre time trial then sets the exact paces you program the running at while you fix the structure. One finds the problem, the other doses the solution.

### Telling the limiters apart: central, peripheral, or tension

When the engine is the problem, it can sit in three different places, and they are trained differently, so it is worth knowing how to tell them apart. Central means delivery: the heart and lungs cannot move enough oxygen, so the whole system runs out at once. Peripheral means use: delivery is fine, but the working muscle cannot extract and burn the oxygen fast enough, so it fails locally while the rest of you feels like it could keep going. Tension is the third, an efficiency problem, where the athlete braces and grips harder than the task needs and chokes off their own blood supply, so they fade from wasted cost rather than a true ceiling.

You tell them apart by where the wheels come off, and you can read most of it without a device. A central limit looks like global breathlessness with the heart rate pinned near max while no single muscle feels like the problem, and it improves when you train delivery, low-end aerobic volume and harder cardiac intervals. A peripheral limit looks like one muscle burning out and failing while breathing is not maxed, and it improves with local muscular endurance, threshold work, and the capillary and mitochondrial work that lets a muscle use what arrives. A tension limit looks like an athlete who is far stronger than the task yet fades fast, grips white-knuckle, holds their breath, and cannot relax what should be relaxed, and it improves with skill, breathing, and position rather than more conditioning. The step test and the benchmark workout are where you watch for this, and the muscle-oxygen tool reads it directly if you have one: muscle oxygen that barely drops points central or to low effort, muscle oxygen that plunges and will not recover points peripheral, and readings that swing with bracing point at tension.

### Finding the zones and pacing without a lab

Most athletes will never see a lactate analyser, and they do not need one. You can anchor the zones two simple ways. If you have power or pace, use the Critical Power test or a time trial and set the zones off that number. If you do not, use heart rate and the talk test from the method library table: easy enough to hold a full conversation is Build, down to short sentences is Endure, cannot talk is Express.

To use heart-rate zones you need a real maximum, and the old formula of two-twenty minus your age is too crude to trust. Find max heart rate with a hard ramp test instead: a

progressive effort on the bike or a hill that builds every minute until you genuinely cannot go harder, and take the highest number you see. Then set your zones as percentages of that, using the ranges in the method library.

Pacing is where heart rate earns its keep and also where it misleads. Heart rate lags. In short, hard intervals it has not caught up by the time the effort is over, so do not chase a heart-rate number on Express work, pace those by power, pace, or feel. Where heart rate is genuinely useful is the easy and moderate end: it keeps your Build work honestly easy, which is the most common thing athletes get wrong, and it lets you watch decoupling on long efforts. If your pace holds steady but your heart rate slowly climbs across a long Build session, that drift is aerobic fatigue showing up, and it is a useful sign that the base still needs work. Use heart rate to police the easy end and to read trends, use power or pace to prescribe the hard end, and use feel to tie it all together.

### **A worked example, the Critical Power test**

As one prescriptive option, two controlled efforts give you the power profile the domains hang off. A short all-out effort of around three minutes shows the powerful, top end. A longer effort of around twelve minutes shows the sustainable, endurant end. Together they give you the athlete's Critical Power and the size of their short-burst reserve, which is what turns the generic zone percentages into this athlete's actual numbers. It is a good test, but it is one tool, not the only one. Choose the test that matches what you are programming.

### **Heart rate, useful but crude**

Heart rate has a place, but be honest about what it is. It drifts with heat, dehydration, caffeine, stress, sleep, and altitude, so the same effort can show very different numbers on different days. As a precise prescription tool it is crude. As a proxy when you have nothing better, it is genuinely useful, especially for keeping easy Build work easy and for watching recovery trends over time. Use it for state and for pacing the easy end. Do not build your precise prescriptions on it if you can measure power or pace instead.

### **Effort by feel, and why it matters for hybrids**

This is where RPE and RIR earn their place, and for a hybrid athlete they are not a fallback, they are a core tool. RPE is how hard a piece feels out of ten. RIR is how many reps you had left on a lift. Both let you prescribe effort for work that has no clean pace or power, and both let you autoregulate, adjusting the work to how the athlete is actually responding that day.

Here is why that is so powerful in a hybrid model. Say you are in a block where building the run is the priority. The running is the limiter, so you prescribe it precisely, exact paces, exact intervals, measured and progressed week to week. But the athlete still lifts and still does some conditioning, and those are not the bias. If you prescribe those precisely too, they fight the running for recovery. So you run them on feel instead. Lift to an RIR rather than a fixed load, condition to an RPE rather than a fixed power. On a day the legs are heavy from running, the athlete naturally backs the lifting off to hold the same RIR, and the running stays protected. On a fresh day, the lifting goes a little harder on its own. The feel-based work undulates with fatigue automatically. That is the whole game: prescribe the limiter tightly, autoregulate everything else by feel, and you bias the limitation while you maintain the rest without grinding the athlete down.

#### **APPLY IT**

For your current goal, name your one limiter. Write that work precisely, with real numbers and a week-to-week progression. Then take everything else in your week and switch it to feel: an RIR on the lifts, an RPE on the conditioning. For the next two weeks, hold the limiter work no matter what, and let the feel-based work rise and fall with how you turn up each day. Notice how much less beaten up you feel while the priority still moves.

## 2. Build the profile and find the limiter

Before any numbers, two questions frame the whole athlete, and they are the most important ones you will ask.

### ***Where did they come from? Where do they want to go?***

Where they came from is their history and their bias. Where they want to go is the goal. Those two set the whole job. From there, three blunt questions do most of the work: where did you come from, what are you good at, and what do you suck at? The last one is usually pointing at the limiter.

### **Expect the archetype trends**

Before you even test, you can predict a lot. Each archetype carries a recognisable profile of what they probably have and what they probably need.

| Archetype        | Likely profile, what they have and need                                                                                                                |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Strength</b>  | Lower cardiac output. Structure built for force, braced rather than breathing. High muscle tension. Aerobic and expression only need to be sufficient. |
| <b>Endurance</b> | High cardiac output. Structure built for breathing and absorbing. Low muscle tension. Strength and expression only need to be sufficient.              |
| <b>Hybrid</b>    | Moderate to high cardiac output. Structure near neutral but always leaning. Moderate tension. The biggest gaps and conflicts, because of interference. |

### **The low-hanging fruit, six things you can change**

Limiters cluster into six levers. Keep what the athlete is, the archetype, separate from how you change them, these six. They often lead into one another: structure / position, strength / tension, cardiac output, skill, VO2max, and expression.

Walk the list against the goal. Are they sufficient in each one, for their sport? The first no that blocks everything downstream is the limiter.

### **The decision-making process**

Finding the limiter is a fixed sequence, run as a loop, not a checklist you do once. Strength gates structure, structure gates skill, skill gates function. You do not move past a gate until it is sufficient.

1. Are they strong enough for their sport?
2. If not, bias strength before going on.
3. Retest strength, then continue.

4. If yes, look at how efficient they are at the task.
5. If not efficient, look at structure.
6. If structure is the limit, train it before going on.
7. Retest the task, then continue.
8. If efficient at the task, train the function itself.
9. If function is not improving, go back to the start.

*The limiter loop. Each gate has to be sufficient before you move to the next.*

## Two worked cases

| Case A, a hybrid athlete                                                                                                                                                                                                                                                               | Case B, a strong CrossFit athlete                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Now: squats 110 kg, runs a slow 5 km.<br/>Wants: squat 140 kg and a strong 5 km.</b>                                                                                                                                                                                                | Now: squats 180 kg, very fit, but fades badly late in long mixed workouts. Wants: hold pace and position deep into a 20-minute workout.                                                                                                                                                 |
| <b>Strong enough for the goal? No. Bias strength first. Once strength is there, check the run. Not an efficient runner, and the cause is structure, they are too extended and braced from heavy lifting to breathe and run well. Train structure, then train the running function.</b> | Strong enough? Easily, yes. Keep strength ticking over. Check the task. Not efficient late in the workout, and the cause is not strength and not structure, it is skill. They waste tension cycling the barbell and moving sloppily under fatigue. Bias skill, then train the function. |

Same process, opposite entries. The hybrid athlete is gated at strength, then structure. The strong CrossFitter is already strong and fit, and is gated at skill. The process tells you which, so you spend the block on the real limiter instead of the obvious one.

## Where is the sport, where is the conditioning

Place two things on the Build, Endure, Express line, separately. First, where the sport's demand actually sits. Second, where the athlete's current conditioning is pointed. The gap between them is the program. A lot of athletes are conditioning a domain their sport never asks for, and that mismatch, once you can see it on the line, is usually the fastest win. Once you have located them: if I am here and I need to be there, I do more work there. Bias the limitation, hold the rest, move on.

### APPLY IT

Draw the domain line and put two dots on it. One for what your goal actually demands, one for where most of your training currently lives. If they are far apart, you have found your block. Move your training toward the demand and stop feeding the domain you are already good at.

## 3. Program

With the limiter found, the build is mechanical. Pick the domain the limiter points to. Pick the method from the library, cyclical or mixed to suit. Choose exercises at the right progression level. Put them on the right day type. Set which domain leads across the block, and let the others sit on a maintenance dose, prescribed precisely if they are the limiter and run on feel if they are not.



## 4. Monitor and autoregulate

Steer by response. Watch RPE and RIR, heart rate variability and resting heart rate, bar speed, and how recovery is trending. Separate the load you put in from the strain it actually causes. Two things sit above everything else and decide whether the plan is working.

| Watch this     | Because                                                                                                                                                            |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tension</b> | It is how you build and how you choke off blood supply and add fatigue. Mismanaged, it is the quiet source of most interference and most connective-tissue injury. |
| <b>Fatigue</b> | Energy is finite, and it is paid from three accounts, training, life, and recovery. The body cannot tell them apart, so you have to.                               |

### What to actually track

Monitoring fails when it gets complicated, so keep it to a few numbers you will actually log every day. A simple system used consistently beats a perfect system you abandon in a week.

- **Every day, in two minutes:** resting heart rate and heart rate variability if you have them, hours and quality of sleep, a soreness score, and an honest one-to-ten for energy and motivation. You are watching the trend, not any single day.
- **Every session:** what you actually did, the loads, paces, or watts, the RPE of the session, the reps in reserve on your key lifts, and whether you hit the prescribed work or had to back off. This is your record of training load and your evidence of whether the limiter is moving.
- **Every week:** a five-minute review. Are the numbers on your limiter improving? Is readiness holding or sliding? That review is what turns logging into decisions.

The signals that tell you to back off are consistent: resting heart rate creeping up, heart rate variability sliding down, bar speed stalling, the same loads feeling harder week to week, sleep falling apart, and motivation draining. One of those is noise. Several together, for several days, is the body telling you to deload.

## 5. Adjust, and test sparingly

Run a method until it stops improving, then change to attack the next limiter. Do not change for variety. Back off on the data, not the calendar, when readiness drops or numbers stall. How long you bias a quality depends on how far it is from sufficient and where you are in the year: bias it hard and briefly when one limiter clearly dominates, gently and longer when several are close together.

Adjusting does not mean formally retesting at the end of every block. Have confidence in the prescriptions and let the training report back: a fixed top set shows the strength trend, easy work at a set heart rate shows the engine trend, and the weekly review reads both without stopping to test. Save the formal retest for the end of a cycle, when a taper has cleared the fatigue and the number is honest, or pull one forward only when the trend stalls and you need to know why. Frequent testing is not free, it adds fatigue and breeds anxiety around the result. The exception is the athlete who wants the feedback and can handle it, who you can test more often by choice, just not by default.

## The deload

A deload is a planned step back so the body can catch up and absorb the work. The trigger is mostly the data, the back-off signals above, but it is also fine to plan one proactively every four to six weeks in a hard block, because you can usually see fatigue coming before it fully arrives. The protocol is simple: drop volume by roughly fifteen to twenty percent and hold the intensity. You cut the amount of work, not the sharpness, so you shed fatigue while keeping the quality and the skill. Alternatively, run a recovery-heavy week, more recovery days, easy Build aerobic, mobility and breathing, and very little hard work. Either way, a deload is not lost time. It is when the previous block's training actually turns into adaptation. Come out of it, re-profile from the trend, and rotate the lead into the next block.

## 6. Recovery is a programmed variable

Recovery, sleep, hydration, and nutrition are not things that happen around the training. They are part of it, prescribed with the same intent as a hard interval. They are also the three non-training accounts that decide whether the training works at all, so you program them on purpose.

- **Sleep.** This is where adaptation actually happens, and it is the cheapest performance gain available. Give it a target, protect it like a session, and judge a hard week partly by whether sleep held. For the time-poor amateur, fixing sleep often beats any change to the program.
- **Nutrition.** Match intake to intent. Enough total energy so the body adapts instead of triaging, enough protein to repair the work, and enough carbohydrate to fuel the priority sessions. Do not try to lose fat and build performance in the same breath.
- **Hydration.** Even mild dehydration raises heart rate and perceived effort and drops output, which costs extra fatigue for the same work. Drink across the day, add fluid and electrolytes around long or sweaty sessions, and do not let a hydration problem look like a fitness problem.
- **Active recovery.** Recovery days, easy Build work, mobility, and breathing are the regeneration engine that makes the expensive days repeatable. They belong in the plan, with a dose, not as an afterthought.

### APPLY IT

Score your last week on the four foundations, sleep, nutrition, hydration, and active recovery, out of ten each. The lowest score is almost certainly costing you more than any tweak to your sets and reps. Fix that one first, and only then go looking for a programming change.

## PART VIII

# Putting It Together

*One athlete, the whole model, from assessment to a written session*

Everything so far has been parts. This is the parts working as one system on a single athlete. Meet Sam, a keen amateur with a demanding office job, two kids, and about five hours a week to train. Sam can squat 130 kg and is strong for a recreational lifter, but gasses badly in anything over ten minutes and dreads running. The goal is a local hybrid event in about twelve weeks: a mix of moderate barbell work and a longer mixed-modal piece with running in it.

## Assess

We test what we will program. A rep-max check confirms Sam is genuinely strong for the goal, well past sufficient. A Critical Power test on the bike gives real numbers to program conditioning against. A short benchmark mixed piece with running in it is the descriptive test, and it is ugly: Sam starts fine and falls apart around the eight-minute mark, breathing ragged, form folding, heart rate pinned. A quick movement screen shows a braced, extended rib position, the classic strong-lifter structure. Resting heart rate and a rough readiness log start on day one.

## Profile and find the limiter

Where did Sam come from? Years of lifting, no conditioning. What is Sam good at? Producing force. What does Sam suck at? Holding output past a few minutes, and running. Run the loop. Strong enough for the goal? Easily, yes, so strength moves to maintenance immediately, there is nothing to chase there. Efficient at the task? No. Is it structure? Partly, the extended, braced position is choking the breathing, that is a real limiter. But the bigger gap is a missing engine: cardiac output and aerobic base were never built, and the running adds an eccentric cost Sam has never adapted to. So the limiter stack is clear. Structure and the aerobic base gate everything, and the engine, biased toward being able to run, is the headline for the block. Strength is support and gets held on a maintenance dose. Put it on the domain line and the picture is stark: the event demands a lot of Endure with an Express edge, and Sam's training has lived almost entirely in short, heavy Express-ish lifting. The gap is the whole program.

## Program, twelve weeks in three blocks

Three blocks walk Sam up the arc, regulate and base, then build, then express, biasing the engine throughout while strength ticks over and structure gets fixed underneath.

| Block                        | Lead / limiter           | Conditioning focus                                                                                   | Strength and structure                                                                                        |
|------------------------------|--------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>1: weeks 1-4<br/>Base</b> | Aerobic base + structure | Build domain, low impact. Aerobic Base and Tension Base on bike, row, ski. Volume progresses weekly. | Strength to maintain: 2 heavy primary sets, twice a week. Daily position and breathing block to get ribs down |

| Block                                   | Lead / limiter           | Conditioning focus                                                                                                                                   | Strength and structure                                                                                                                                             |
|-----------------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                         |                          | Running introduced tiny and easy, to adapt the tissue.                                                                                               | over pelvis.                                                                                                                                                       |
| <b>2: weeks 5–8<br/>Build</b>           | Endure engine            | Endure domain. Edge Intervals and Recoverable Intervals on the bike, Loaded Mixed pieces, running volume rising slowly. This is the main block.      | Strength still maintenance. Structure work continues but shifts into warm-ups as it becomes available. Add isometrics for the tendons taking the new running load. |
| <b>3: weeks 9–12<br/>Express + peak</b> | Task-specific expression | Endure held, plus Express and mixed pieces that look like the event: running into barbell work, paced then pushed. Skill under fatigue is the focus. | Strength touched once a week to stay sharp. Structure now mostly in warm-ups. Deload week 11, event week 12.                                                       |

## A week from block two

This is Sam's actual week in the middle of the build, five training days, engine leading, everything else held. It follows the day-type engine: stimulate then develop, hard work spaced, recovery kept genuinely easy.

| Day        | Type           | Main work                                                                                                             | Conditioning                         | Layer                    |
|------------|----------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------|--------------------------|
| <b>Mon</b> | Dev            | Warm-up with position block. Back squat 3x3 heavy (maintain). Then the engine: Edge Intervals, 2x15 min bike @ RPE 7. | Endure, cyclical.                    | Mental: express          |
| <b>Tue</b> | Stim           | Upper push/pull, held at RIR 3. Trunk and isometric calf work for the tendons.                                        | Easy 30 min Build bike.              | Regulate                 |
| <b>Wed</b> | Recovery       | Mobility, breathing, light sled drags. Nothing heavy.                                                                 | Build, 40 min easy + short easy run. | Anchor                   |
| <b>Thu</b> | Dev            | Warm-up. Trap-bar deadlift 3x3 (maintain). Loaded Mixed piece: 18 min AMRAP, ski / carry / lunge @ RPE 7–8.           | Endure, mixed.                       | Mental: express          |
| <b>Fri</b> | Off / regulate | Rest. Evening breathing and a short visualization of the event.                                                       | –                                    | Spiritual                |
| <b>Sat</b> | Stim → long    | Progressive run building easy to moderate, then a longer easy Build ride.                                             | Build → Endure.                      | Spiritual: long practice |
| <b>Sun</b> | Off            | Rest and reflection.                                                                                                  | –                                    | –                        |

Read the model in it. The engine is the lead, so conditioning carries the volume and progresses week to week. Every lift says maintain, two or three heavy sets to hold the strength Sam already has, never enough to chase more, because strength is support here. The runs start tiny and grow slowly so the tendons adapt instead of tearing, exactly the connective-tissue caution the model insists on. Position and breathing work is daily early, then folds into warm-ups as it becomes available. The mental and spiritual layers are mapped onto the day types, not bolted on: express work on the hard days, regulate and anchor on the easy ones.

## One session, written out

Here is Monday in full, so the whole model is visible in a single session, from warm-up to cooldown, running the level-one-to-level-three ramp.

| Phase                    | What Sam actually does                                                                                                                                     |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Warm-up: raise</b>    | 5 min easy bike, Build pace, just getting warm.                                                                                                            |
| <b>Warm-up: position</b> | 90/90 breathing and a dead bug series, long exhales, ribs down over pelvis. Then a few stacked goblet-squat holds to own the position under a little load. |
| <b>Warm-up: ramp</b>     | Empty bar to working weight on the squat over 3-4 building sets, last one close to the working load.                                                       |
| <b>Priority strength</b> | Back squat 3x3 at a heavy but maintenance load, RIR 2-3. Crisp, fast reps. Strength is support, so this is quality, not a grind.                           |
| <b>Priority engine</b>   | Edge Intervals: 2x15 min on the bike at RPE 7, around 85% max HR, 5 min easy between. This is the headline, the limiter, the reason for the session.       |
| <b>Cooldown</b>          | 5 min easy spin bringing the heart rate down, then 2-3 min of slow breathing, long exhales, to start recovery and train down-regulation.                   |

Notice the order: fresh, high-quality strength first, then the engine work that carries the fatigue, then a cooldown that starts recovery and doubles as mental regulate training. One session, all three layers, the whole model in miniature.

## Monitor and adjust

Across the block Sam logs resting heart rate, sleep, and a readiness score each morning, and the session RPE and RIR each day. The engine numbers climb, the same bike intervals sit at a lower heart rate, the benchmark piece stops falling apart at eight minutes. In week 7 resting heart rate creeps up for four days and readiness slides, so the Thursday piece is cut short and Friday made fully off, load backed off on the data, not the calendar. A planned deload in week 11 sheds the fatigue before the event. Strength, checked once, has barely moved, exactly as intended, because it was never the job. The engine, the limiter, is transformed. That is the model working: bias the limitation, hold the rest, steer by response, aim it at the goal.

### APPLY IT

Run yourself through Sam's process. Test what you will program, answer the three blunt questions, run the limiter loop, and put two dots on the domain line. Then write your own

twelve weeks in three blocks, biasing your one limiter, holding everything else on a maintenance dose, and mapping the mental and spiritual work onto your day types. You now have the whole system in your hands.

## PART IX

# The Full Chapter Map

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## *How the framework maps onto the chapters of the book*

For readers who want the linear route, here is how the whole model lays out chapter by chapter.

- **Chapter 1, The Philosophy.** The three layers, the shared arc, the shared nervous system, and the two refrains. Bias the limitation but train the full system. Build the base, earn the expression, aim it higher.
- **Chapter 2, Defining Fitness.** The Fitness Compass, descriptive and prescriptive qualities, structure and function, skill and tension.
- **Chapter 3, The Engine.** The oxygen-led model, the cascade of lungs, heart, and muscle, fibre types, the shared nervous system, and the levers on each link.
- **Chapter 4, The Domains.** Build, Endure, Express, the two thresholds, reading the domains in halves, the five dials, and the contraction axis.
- **Chapter 5, Building Capacity.** Loading connective tissue, hypertrophy and strength inside the domains, combining methods, and interference.
- **Chapter 6, The Programming Engine.** The movement progression, the method library, strength programming, session and week structure, and building the program.
- **Chapter 7, The Mental Layer.** Regulate, training outside as the oldest regulator, build, express, sustaining, and the failure ladder.
- **Chapter 8, The Spiritual Layer.** Anchor, aim, embody, and the goal ladder that turns the aim into one repeatable habit.
- **Chapter 9, Integration.** Three layers, one nervous system, one calendar.
- **Chapter 10, Running It End to End.** Assess, profile, program, monitor, adjust, and recovery as a programmed variable.
- **Chapter 11, Putting It Together.** The full worked example, from assessment to a single written session.



# Quick Reference

*The whole system on a few pages*

## The loop

Test → profile → find the limiter → program → train → monitor → adjust → retest. Bias the limitation, but train the full system.

## The domains at a glance

| Domain         | RPE  | % max HR | Talk test         | Job, and recovery cost                                           |
|----------------|------|----------|-------------------|------------------------------------------------------------------|
| <b>Build</b>   | 3–5  | 50–75%   | Full conversation | Deliver. Build the engine cheaply. Repeatable daily.             |
| <b>Endure</b>  | 6–8  | 75–90%   | Short sentences   | Hold output at the edge. 24–48 h to repeat.                      |
| <b>Express</b> | 9–10 | 90–100%  | Cannot talk       | Produce maximal output. ~72 h to repeat.                         |
| <b>Z7</b>      | Max  | n/a      | n/a               | Pure speed and power. Short bouts, full rest, low systemic cost. |

## The shades within the domains

Each domain splits into a low and a high half. Use the split as a progression (own the low half first) and as an adaptation target (each half trains a different quality). Z7 sits above the six as the alactic top.

| Shade               | RPE  | % max HR | Trains, and recovery                                                                      |
|---------------------|------|----------|-------------------------------------------------------------------------------------------|
| <b>Build low</b>    | 3–4  | 50–65%   | Delivery base: cardiac output, capillaries, fat oxidation. No recovery cost, train daily. |
| <b>Build high</b>   | 4–5  | 65–75%   | Base worked harder, muscular endurance. Low cost, back-to-back fine.                      |
| <b>Endure low</b>   | 6–7  | 75–85%   | Pacing and heart-rate recoverability. Inside 24 h, not smashed daily.                     |
| <b>Endure high</b>  | 7–8  | 85–90%   | Threshold, raising the sustainable ceiling. ~48 h.                                        |
| <b>Express low</b>  | 9    | 90–95%   | The aerobic ceiling, VO <sub>2</sub> , central delivery. ~72 h.                           |
| <b>Express high</b> | 9–10 | 95–100%  | Lactate tolerance and glycolytic power. ≥72 h, the biggest recovery bill you program.     |
| <b>Z7</b>           | Max  | n/a      | Alactic power and pure speed. Low systemic cost, full rest between bouts.                 |

## The method map

Every method carries a domain and a shade, so the library and the shade table speak one language. Spring Prep and the Z7 repeats sit off the metabolic line, on the tissue and neuromuscular track.

| Method                       | Domain           | Shade                        | Mode / note                                      |
|------------------------------|------------------|------------------------------|--------------------------------------------------|
| <b>Aerobic Base</b>          | Build            | Low → high                   | Cyclical or mixed                                |
| <b>Tension Base</b>          | Build            | Low → high                   | Cyclical                                         |
| <b>Spring Prep</b>           | Build-adjacent   | Tissue (not metabolic)       | Tissue / neuromuscular, connective-tissue ladder |
| <b>Recoverable Intervals</b> | Endure           | Low                          | Cyclical                                         |
| <b>Carry and Brace</b>       | Endure           | Low → high                   | Mixed                                            |
| <b>Edge Intervals</b>        | Endure           | High                         | Cyclical                                         |
| <b>Grind</b>                 | Endure           | High                         | Cyclical                                         |
| <b>Loaded Mixed</b>          | Endure           | High                         | Mixed                                            |
| <b>Lactic Pieces</b>         | Endure → Express | High (top), into Express low | Mixed, long rest                                 |
| <b>Peak Intervals</b>        | Express          | Low                          | Cyclical                                         |
| <b>Heavy Mixed</b>           | Express          | High                         | Mixed                                            |
| <b>Power Repeats</b>         | Z7               | — (alactic top)              | Cyclical, full rest                              |
| <b>Reactive Repeats</b>      | Z7               | — (alactic top)              | Mixed / neuromuscular, full rest                 |
| <b>Engine Builder</b>        | Cross-domain     | Endure → Express             | Bridge / mixed                                   |

## The day types

| Day                | Stress   | RPE  | Role                                                                  |
|--------------------|----------|------|-----------------------------------------------------------------------|
| <b>Development</b> | High     | 9-10 | The expensive day. Drives adaptation. Mental express work lives here. |
| <b>Stimulation</b> | Moderate | 7-8  | The priming day. Sets up development. Mental build work.              |
| <b>Recovery</b>    | Low      | ≤6   | Regeneration. Easy Build work. Regulate and anchor work.              |

## The decision process

Strong enough for the sport? If not, bias strength. If yes, efficient at the task? If not, check structure and fix it. If structure is fine, train the function itself. If function will not improve, go back to the start. Each gate must be sufficient before the next.

## The refrains

***Bias the limitation, but train the full system.***

***Build the base, earn the expression, aim it higher.***

# Glossary

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*Every term, in one place*

**Build, Endure, Express.** The three training domains across the line of demand, set by two thresholds. Build delivers cheaply, Endure holds output at the edge, Express produces maximal output. Also a low-to-high tension and skill gradient.

**Shades (low / high).** Each domain splits into a low and a high half, six metabolic shades in total, defined by lactate and oxygen. The split does two jobs: it is a progression (build the low half before loading the high half) and it aims at a different adaptation (delivery, threshold, VO<sub>2</sub>, lactate tolerance). Steer by RPE; use the effort and heart-rate figures to pin a shade when biasing hard. See Reading the domains in halves.

**Z7.** The pure alactic top, a seventh shade that sits above the metabolic six, not inside them. Short maximal bouts with full recovery, lactate fully cleared between reps, low systemic cost despite the intensity. Home of Power Repeats and Reactive Repeats.

**T1 and T2.** The two thresholds. T1 sits between Z2 and Z3, where lactate first rises above baseline. T2 sits between Z4 and Z5, the top of metabolic steady state, above which lactate accumulates faster than you can clear it.

**Critical Power / Critical Speed (CP / CS).** The hardest effort you can sustain for a long time without falling apart. Anchors the zones and separates sustainable work from the short reserve above it.

**RPE.** Rate of perceived exertion, how hard a whole piece feels out of ten. The default dial for conditioning and for maintaining a quality.

**RIR.** Reps in reserve, how many reps you had left on a lift. The default dial for lifting and for autoregulating strength work by feel.

**Cascade (uptake, transport, expression).** The oxygen supply chain: lungs take oxygen in, heart moves it, muscle uses it. Names the links where a limiter can sit. Not the same words as the training domains.

**Central, peripheral, tension.** The three places an engine limit can sit. Central is delivery (heart and lungs), peripheral is use (the muscle extracting oxygen), tension is wasted efficiency (bracing that chokes off blood supply). Each is trained differently.

**Structure and function.** Structure is the athlete's architecture, chiefly the rib-cage-to-pelvis position and the breathing it allows. Function is the combination of energy systems and muscle fibres organised for a task. Structure dictates function; function over time reshapes structure.

**The contraction line.** Concentric (own the shortening), isometric (hold position), tempo eccentric (control the lowering), reactive (release at speed), aligned to the domains. Each stage earns the next.

**The five dials.** The five ways to put a number on effort: RPE and RIR (feel, for maintaining and mixed work), and watts, run pace, and percent of max (precision, for biasing a specific quality).

**Headline and support.** For any athlete, one quality is the headline you keep building and the other is the support that only needs to be sufficient. Past sufficient, support work is wasted training.

**The sufficiency principle.** Several qualities only need to be good enough for the sport, not maximal. Satisfy the requirement, then spend the budget on the real limiter.

**The three fatigue accounts.** Training, life, and recovery. The body pays fatigue from all three and cannot tell them apart, so total load, not just training, decides what an athlete can handle.

**Day types.** Development (high stress, drives adaptation), stimulation (moderate, primes), recovery (low, regenerates). The week and all three layers map onto them.

**Bias the limitation, but train the full system.** Give the limiting quality the lead, hold everything else on a maintenance dose, then retest and rotate. The core operating discipline.

**Build the base, earn the expression, aim it higher.** The shared arc of all three layers: regulate, build, express, pointed at something above the self.